Almost sure continuity of the LQG metric

Abstract

We prove that the median-normalized Liouville quantum gravity metrics associated with a single whole-plane Gaussian free field admit a modification which is almost surely continuous in the parameter $\xi \in (0, \infty)$ for the topology of lower semicontinuous functions. The proof is based on a quantitative comparison of metrics with nearby parameters. Smooth changes to the field produce local shortcuts, which are amplified using annular crossings. Summable comparisons on nested parameter grids then give the continuous modification.

1 Introduction

Let h be a whole-plane Gaussian free field, with its additive constant chosen so that its average over $\partial B_1(0)$ is zero. For $\xi > 0$, let D_h^ξ be the Liouville first passage percolation limit of [Kav25, Theorem 1.4], with LFPP distances normalized by their crossing medians. This metric exists for every $\xi > 0$ and is unique up to a deterministic multiplicative constant [DG22]. In the supercritical regime it is lower semicontinuous and may take the value ∞.

For each $\xi > 0$, let $\beta(\xi)$ be the median of $D_h^\xi(0, 1)$. The median exists by [Kav25, Lemma 10.1], and its uniqueness will also follow from the proof below. Set

$$\widetilde{D}_h^\xi = \beta(\xi)^{-1} D_h^\xi. \tag{1.1}$$

All metrics in this paper are coupled using the same field h.

We use the following topology, as in [Kav25, Definition 1.3]. A sequence of lower semicontinuous functions $f_n : \mathbb{C}^2 \to [0, \infty]$ converges to f if

(i) for every $(z_n, w_n) \to (z, w)$, $f(z, w) \le \liminf_n f_n(z_n, w_n)$;

(ii) for every (z, w), there is a sequence $(z_n, w_n) \to (z, w)$ for which $f_n(z_n, w_n) \to f(z, w)$.

We call this the *lsc topology*. Let $\mathcal{M}$ be the space of lower semicontinuous extended metrics on $\mathbb{C}$, equipped with this topology. Here an extended metric is symmetric, vanishes precisely on the diagonal, satisfies the triangle inequality, and may equal ∞ off the diagonal.

Theorem 1.1. *The family $\{\widetilde{D}_h^\xi\}_{\xi>0}$ admits a jointly measurable modification $\{\widehat{D}_h^\xi\}_{\xi>0}$ such that, almost surely,*

$$(0, \infty) \ni \xi \longmapsto \widehat{D}_h^\xi \in \mathcal{M}$$

is continuous.

In particular, for each deterministic sequence $\xi_n \to \xi > 0$, $\widetilde{D}_h^{\xi_n} \to \widetilde{D}_h^\xi$ almost surely in the lsc topology. The modification in Theorem 1.1 is needed because separately chosen versions can differ on parameter-dependent null sets.

Kavvadias [Kav25, Theorem 1.6] proves tightness of the family under parameter convergence and identifies subsequential limits, assuming that the limit satisfies the triangle inequality almost surely. Our proof gives quantitative comparisons on the original field probability space and proves the triangle inequality at every parameter of the continuous modification.

1.1 The comparison estimate

It is convenient to work first with the normalization

$$m(\xi) = \exp \mathbb{E} \log D_h^\xi(0,1), \qquad D^\xi = m(\xi)^{-1} D_h^\xi. \tag{1.2}$$

The moment estimates below imply that m is finite, positive, and bounded above and below on compact parameter intervals. No continuity of m is assumed. For an extended metric D and $r > 0$, write

$$D_r(x,y) = \inf\{D(u,v) : u \in \overline{B}_r(x),\ v \in \overline{B}_r(y)\}. \tag{1.3}$$

Define $T_0(s) = s$ and $T_{j+1}(s) = \exp(T_j(s))$.

Proposition 1.2. *Fix a compact interval $I \Subset (0,\infty)$. There are constants $c, C, L > 0$ and an integer $d \geq 1$, depending only on I, with the following property. Suppose $\xi, \eta \in I$, $0 < \delta < c$, and $n \in \mathbb{N}$ satisfy*

$$|\xi - \eta| \leq T_d(\delta^{-C})^{-1}, \qquad n \geq C\delta^{-L}, \qquad |\xi - \eta| n \leq c\delta. \tag{1.4}$$

Set $r = \exp(-c\delta^L n)$. For $1 \leq R \leq \exp(c\delta^L n)$, except on an event of probability at most

$$C(R+1)^C \delta^{-L} \exp(-c\delta^L n), \tag{1.5}$$

we have, simultaneously for every $x, y \in B_R(0)$,

$$D_r^\eta(x,y) \leq (1+\delta) D^\xi(x,y), \qquad D_r^\xi(x,y) \leq (1+\delta) D^\eta(x,y). \tag{1.6}$$

The dependence on δ in (1.4) is needed to control all edges of a sequence of increasingly fine parameter grids. A qualitative comparison for each fixed accuracy would not suffice for this purpose. The endpoint displacement in (1.6) is also retained when passing to the limit: it allows us to join recovered paths and prove the triangle inequality at exceptional parameters.

The proof of Theorem 1.1 from Proposition 1.2 is given first. We then prove the comparison estimate. The analytic part consists of short-loop estimates and an annular amplification argument. The geometric part produces local shortcuts by subtracting smooth functions from the field and uses a finite counting argument to improve the comparison coefficient to one. The compact-uniform LFPP estimates and the detailed planar construction are included in the appendices.

1.2 Notation and conventions

We write $B_r(z)$ for a Euclidean open ball and $A_{r,R}(z) = B_R(z) \setminus \overline{B}_r(z)$. The circle average of h over $\partial B_r(z)$ is denoted by $h_r(z)$. Put $q(\xi) = \xi Q(\xi)$, so the physical metric unit at scale r about z is $r^{q(\xi)} e^{\xi h_r(z)}$. The function q is locally Lipschitz, as proved in the uniform estimates below.

For a set U, the internal metric $D(\cdot,\cdot;U)$ is the infimum of the D-lengths of paths contained in U. When U is a closed ball, paths may use its boundary. We write $D(\text{across } A)$ and $D(\text{around } A)$ for the crossing and separating-loop distances of an annulus. Circle measures are Euclidean arclengths.

Throughout a proof with two parameters, $D = D^\xi$, $E = D^\eta$, and $\tau = |\xi - \eta|$. Constants may depend on a fixed compact interval I and on the fixed Euclidean domains under consideration. They are uniform in $\xi, \eta \in I$. Constants denoted by c, C may change from line to line.

We use the fixed-parameter locality, Weyl scaling, coordinate scaling, and geodesic properties from [DG22]. When a function f ranges over a class of perturbations, the shifted metrics are defined pathwise by Weyl weighting. For each deterministic countable collection of perturbations these agree simultaneously with the corresponding canonical metrics. All events are interpreted in the completion of the Gaussian probability space.

2 Proof of almost-sure continuity

Throughout this section, $I = [a, b] \Subset (0, \infty)$ is a compact interval with rational endpoints. We use

$$m(\xi) = \exp \mathbb{E} \log D_h^\xi(0, 1), \qquad D^\xi = m(\xi)^{-1} D_h^\xi.$$

Proposition 3.1 implies that m is finite, positive, and bounded above and below on I. All the estimates of that proposition therefore also hold for D^ξ.

2.1 A complete metric for lower semicontinuous convergence

Let $\mathcal{F}$ be the space of symmetric, nonnegative lower semicontinuous functions on $X = \mathbb{C}^2$ which vanish on the diagonal. We equip X with $d_X((x, y), (u, v)) = |x - u| + |y - v|$. For $f \in \mathcal{F}$, set

$$T_m f(z) = \inf_{w \in X} \{f(w) + m d_X(z, w)\}, \qquad \rho(f, g) = \sum_{m,R \geq 1} 2^{-m-R} (1 \wedge \|T_m f - T_m g\|_{K_R}), \tag{2.1}$$

where $K_R = \overline{B}_R(0)^2$ and $\| \cdot \|_{K_R}$ is the uniform norm.

Lemma 2.1. *The metric ρ makes $\mathcal{F}$ a complete separable metric space. Its convergence is exactly the lower semicontinuous convergence of Definition 1.3 of [Kav25].*

Proof. Each $T_m f$ is nonnegative and m-Lipschitz, vanishes on the diagonal, and satisfies $T_m f(x, y) \leq m|x - y|$. Moreover, $T_m f \uparrow f$. Indeed, if $T_m f(z)$ stays bounded, its approximate minimizers converge to z, and lower semicontinuity gives the reverse inequality to $T_m f \leq f$. This also proves that ρ separates points.

Suppose $f_n \to f$ in the lower semicontinuous topology. Minimizers of $T_m f_n(z)$ lie in a common compact set, by nonnegativity and the diagonal competitor. Taking a convergent subsequence and using the liminf condition gives $T_m f(z) \leq \liminf_n T_m f_n(z)$. A recovery sequence for a minimizer of $T_m f(z)$ gives the reverse limsup inequality. The common Lipschitz bound upgrades pointwise convergence to local uniform convergence, hence to convergence in ρ.

Conversely, suppose the envelopes converge locally uniformly. For $z_n \to z$ and every m,

$$\liminf_n f_n(z_n) \geq \lim_n T_m f_n(z_n) = T_m f(z).$$

Letting $m \to \infty$ proves the liminf condition. If $f(z) < \infty$, choose integers $m_n \to \infty$ sufficiently slowly that $|T_{m_n} f_n(z) - T_{m_n} f(z)| \leq m_n^{-1}$. Approximate minimizers give

$$f_n(z_n) + m_n d_X(z, z_n) \leq f(z) + 2m_n^{-1}.$$

Thus $z_n \to z$ and $\limsup_n f_n(z_n) \leq f(z)$. If $f(z) = \infty$, the constant sequence is a recovery sequence by the liminf condition.

For completeness, let (f_n) be ρ-Cauchy and write g_m for the local uniform limit of $T_m f_n$. The functions g_m increase with m. Infimal convolution satisfies

$$T_\ell T_m f = T_{\min\{\ell, m\}} f.$$

This follows by expanding the infima and applying the triangle inequality; either endpoint can be chosen as the intermediate point. Each T_ℓ preserves local uniform convergence of nonnegative functions vanishing on the diagonal, since its minimizers stay in a common compact set. Consequently $T_\ell g_m = g_{\min\{\ell, m\}}$.

Set $f = \sup_m g_m \in \mathcal{F}$. Then $T_\ell f \geq g_\ell$. For the converse, choose minimizers w_m for $T_\ell g_m(z) = g_\ell(z)$, $m \geq \ell$. They remain in a compact ball. Along a convergent subsequence, with limit w, continuity and monotonicity of the g_m imply, for every fixed j, $g_j(w) + \ell d_X(z,w) \leq g_\ell(z)$. Taking the supremum over j proves $T_\ell f \leq g_\ell$. Hence $f_n \to f$ in ρ. Finally, the envelope map embeds $\mathcal{F}$ in the countable product of the separable spaces $C(K_R)$, so $\mathcal{F}$ is separable. □

Recall that $E \preceq_{\delta,r} D$ means that each finite $D(x,y)$ has recovery endpoints x', y' within r of x, y, respectively, with $E(x',y') \leq (1+\delta)D(x,y) + r$.

Lemma 2.2. *If the reciprocal comparisons $E \preceq_{\delta,r} D$ and $D \preceq_{\delta,r} E$ hold on $\overline{B}_{3S}(0)$, then*

$$\rho(D,E) \leq 2^{-M} + 2^{-S} + 2MS\delta + (2M+1)r, \qquad M, S \in \mathbb{N}. \tag{2.2}$$

Proof. For $(x,y) \in K_R$ and $m \leq M$, a minimizer (u,v) for $T_m D(x,y)$ has cost at most $m|x-y| \leq 2mR$. Thus $u, v \in \overline{B}_{3R}$ and $D(u,v) \leq 2mR$. Using its recovery endpoints as a competitor for $T_m E$ increases the cost by at most $2mR\delta + (2m+1)r$. Reverse the roles of D, E, then sum for $m \leq M$, $R \leq S$ and bound the remaining weights. □

2.2 A continuous modification

Use the constants d, C, c, L from Proposition 1.2, and put $t(\delta) = T_d(\delta^{-C})^{-1}$. Write $T_0(s) = s$ and $T_{j+1}(s) = \exp(T_j(s))$. Choose an integer $D \geq d+2$ and B sufficiently large, and set

$$\begin{aligned} &\delta_j = 2^{-j-10}(j+1)^{-2}, \qquad k_j = \lceil T_D(2^{Bj}) \rceil, \qquad h_j = (b-a)2^{-k_j}, \\ &Q_j = \{a + \ell h_j : 0 \leq \ell \leq 2^{k_j}\}, \qquad R_j = j + 10, \\ &n_j = \lfloor c\delta_j/(2h_j) \rfloor, \qquad A_j = c\delta_j^L n_j, \qquad r_j = e^{-A_j}, \qquad s_j = \sum_{i \geq j} r_i. \end{aligned} \tag{2.3}$$

The grids are nested. For sufficiently large j, the scale restrictions in Proposition 1.2 hold whenever $|\xi - \eta| \leq h_j$. The fixed tower bound for t also gives

$$\frac{A_j}{k_{j+1} + j} \longrightarrow \infty, \qquad r_j \leq s_j \leq 2r_j, \qquad \sum_j 2^{k_{j+1}} e^{-qA_j} < \infty \quad (q > 0). \tag{2.4}$$

To check these claims, note that $A_j \asymp \delta_j^{L+1} 2^{k_j}$, whereas $k_{j+1} = \lceil T_D(2^B 2^{Bj}) \rceil$. An additional exponential dominates a fixed-factor increase in the bottom argument of a fixed tower; repeated logarithms prove the first claim and $A_{j+1}/A_j \to \infty$. These imply the other two claims. The same comparison proves $h_j \leq t(\delta_j)$.

Lemma 2.3. *There is a jointly measurable family $(F_h^\xi)_{\xi \in I}$ in $\mathcal{F}$ whose trajectories are almost surely continuous and such that $F_h^\xi = D^\xi$ almost surely for each deterministic $\xi \in I$.*

Proof. Intersect the fixed-parameter probability-one events over $\bigcup_j Q_j$. For each $u \in Q_{j+1}$ compare u with its left projection in Q_j; also compare adjacent points of Q_j. There are at most $C2^{k_{j+1}}$ such pairs, each at distance at most h_j. Proposition 1.2, (2.4), and Borel–Cantelli imply that, almost surely, all their reciprocal comparisons hold on B_{R_j} for every sufficiently large j. By Lemma 2.2, with $M = S = \lfloor j/4 \rfloor$,

$$\rho(D^u, D^v) \leq e_j, \qquad e_j = C(2^{-j/4} + j^2\delta_j + jr_j), \qquad \sum_j e_j < \infty. \tag{2.5}$$

Let $q_j(\xi)$ be the left projection of ξ onto Q_j, with $q_j(b) = b$. These projections form parent chains. Completeness gives simultaneous limits

$$F_h^\xi = \lim_{j\to\infty} D^{q_j(\xi)}, \qquad \sup_{\xi\in I} \rho(F_h^\xi, D^{q_j(\xi)}) \le \sum_{i\ge j} e_i.$$

If $|\xi - \eta| < h_j$, their level-j projections coincide or are adjacent. Therefore

$$\rho(F_h^\xi, F_h^\eta) \le 2\sum_{i\ge j} e_i + e_j,$$

which proves continuity. Assign a fixed element of $\mathcal{F}$ on the null exceptional event. The approximating step functions are jointly measurable in (h, ξ), so their limit is jointly measurable.

For every deterministic sequence $\xi_n \to \xi$ in I,

$$\rho(D^{\xi_n}, D^\xi) \longrightarrow 0 \quad \text{in probability.} \tag{2.6}$$

Indeed, fix M, S and a small $\delta > 0$, and use Proposition 1.2 at depth $\lfloor c\delta/(2|\xi_n - \xi|)\rfloor$; equal parameters need no comparison. Eventually all its restrictions hold, the failure probability tends to zero, and the recovery radius tends to zero. Equation (2.2) then proves the claim by first taking M, S large and then δ small. Apply this to $q_j(\xi)$. Uniqueness of limits in probability identifies its almost-sure limit as D^ξ, proving the modification identity. □

2.3 Metricity of the modification

Lemma 2.4. *Almost surely, every F_h^ξ, $\xi \in I$, is an extended metric.*

Proof. We retain the reciprocal endpoint comparisons used above. For each fixed spatial disk B_R, the inverse-Hölder estimate in Proposition 3.1, a union bound over Q_j, and Borel–Cantelli give, eventually, for all $\theta \in Q_j$,

$$D^\theta(u, v) \ge H_j^{-1}(|u - v| \wedge 1)^\chi \quad (u, v \in B_R), \qquad H_j = e^{C_R k_j}. \tag{2.7}$$

Choose C_R large enough for the polynomial tail of the inverse-Hölder constant. Likewise, Corollary 3.4 and (2.4) imply that, eventually, every $\theta \in Q_j$ and $y \in B_R$ have $J_j \ge c \log s_j^{-1}$ disjoint annuli surrounding $B_{2s_j}(y)$, contained in $B_{s_j^{1/2}}(y) \setminus \overline{B}_{2s_j}(y)$, on which

$$D^\theta(\text{around } A) \le C D^\theta(\text{across } A). \tag{2.8}$$

Indeed, the failure probability is at most $C_R 2^{k_j} s_j^q$, summable for any fixed $q > 0$. Intersect these events over integer R.

Work on this one probability-one event and fix any $\xi \in I$. Write $D_j = D^{q_j(\xi)}$, $f = F_h^\xi$, and $P_j = \prod_{i\ge j}(1 + \delta_i) \to 1$. For every $f(x, y) < \infty$, backward composition of the comparisons gives points x_j, y_j with

$$|x_j - x| \vee |y_j - y| \le s_j, \qquad D_j(x_j, y_j) \le P_j(f(x, y) + s_j). \tag{2.9}$$

To justify this, take a recovery sequence at levels $N \to \infty$ and compose from N down to j. Total endpoint displacement is at most s_j, and total additive error is at most $P_j s_j$. At fixed j, compactness of the endpoints and lower semicontinuity of D_j give (2.9). For sufficiently large j, all endpoints of the composition lie in its comparison windows.

If $x \neq y$ and $f(x,y) = 0$, equations (2.7) and (2.9) contradict $H_j P_j s_j \to 0$, which follows from (2.4). Thus f is positive off the diagonal.

For the triangle inequality, let x, y, z be distinct with $f(x,y) + f(y,z) < \infty$. Apply (2.9) to both legs, obtaining endpoints (x_j, y_j) and (y'_j, z_j). Choose D_j-paths of lengths

$$a_j \leq P_j(f(x,y) + s_j) + j^{-1}, \qquad b_j \leq P_j(f(y,z) + s_j) + j^{-1}.$$

They cross all the J_j annuli around y for large j. Disjointness implies that the sum of their across-distances is at most $\min\{a_j, b_j\}$. One annulus therefore has a separating loop of length at most $2C \min\{a_j, b_j\}/J_j$. Both paths meet this loop. Concatenating the appropriate path portions with a loop arc gives

$$D_j(x_j, z_j) \leq a_j + b_j + 2C \min\{a_j, b_j\}/J_j.$$

The liminf condition, $P_j \to 1$, $s_j \to 0$, and $J_j \to \infty$ give $f(x,z) \leq f(x,y) + f(y,z)$. Coincident points or an infinite right-hand side are immediate. This deterministic argument applies to every ξ and every triple on the event already chosen. □

2.4 The median normalization

Lemma 2.5. *The function $b(\xi) = \beta(\xi)/m(\xi)$ is positive and continuous on $(0, \infty)$.*

Proof. For fixed ξ, the law of $D_h^\xi(0,1)$ is atomless by [Kav25, Lemma 10.1]. We first check that it has full support in $(0, \infty)$, so its median is unique.

Choose a smooth compactly supported function g with $g_1(0) = 0$ which equals -1 on a narrow open tube U joining 0 to 1. Such a function is obtained by correcting its circle average with a bump on a disjoint arc of the unit circle. Proposition 3.1 gives $D_h^\xi(0,1;U) < \infty$ almost surely, and Weyl scaling gives

$$D_{h+tg}^\xi(0,1) \leq e^{-\xi t} D_h^\xi(0,1;U) \longrightarrow 0.$$

Cameron–Martin equivalence, applied to integer t, implies $\mathbb{P}[D_h^\xi(0,1) < a] > 0$ for every $a > 0$.

Next choose a nonnegative smooth function f supported in $B_{1/2}$, equal to one on $A_{1/8,1/4}$. It preserves the field pin. Every path from 0 to 1 crosses this annulus, so for $t \geq 0$,

$$D_{h+tf}^\xi(0,1) \geq D_h^\xi(0,1) + (e^{\xi t} - 1) D_h^\xi(\text{across } A_{1/8,1/4}).$$

The last crossing distance is positive. Bounded Weyl scaling also makes the left side continuous in t. Starting on the positive probability event $D_h^\xi(0,1) < a$, some rational $t \geq 0$ therefore places it in any prescribed interval (a, b). A countable union and Cameron–Martin equivalence prove full support. Only countably many shift identities are needed; the Weyl length formula supplies their continuous interpolation.

We next prove, for deterministic $\xi_n \to \xi$,

$$D^{\xi_n}(0,1) \longrightarrow D^\xi(0,1) \quad \text{in probability.} \tag{2.10}$$

Terms with $\xi_n = \xi$ are immediate. Fix small $\delta, v > 0$ and apply Proposition 1.2 at depth $\lfloor c\delta/(2|\xi_n - \xi|) \rfloor$. Its failure probability and recovery radius r_n tend to zero. Lemma 3.2, applied to both metrics at $0, 1$, joins each fixed endpoint to a loop enclosing its r_n-ball, at total cost at most v, except with probability at most $Cv^{-p} r_n^s$. For large n the two loop neighborhoods are disjoint. A path between recovery endpoints meets both loops, so truncating it and adding the access paths and loop arcs gives

$$D^{\xi_n}(0,1) \leq (1+\delta) D^\xi(0,1) + r_n + 2v,$$
$$D^\xi(0,1) \leq (1+\delta) D^{\xi_n}(0,1) + r_n + 2v.$$

These inequalities hold with probability tending to one. The uniform point moments make the distances tight. Letting $\delta, v \to 0$ proves (2.10).

The variable $D^{\xi}(0,1)$ consequently has a continuous, strictly increasing distribution function and unique median $b(\xi)$. At two thresholds on opposite sides of this median the distribution function is respectively below and above $1/2$. Equation (2.10) gives the same inequalities for $D^{\xi_n}(0,1)$ for large n, trapping its median between those thresholds. Hence $b(\xi_n) \to b(\xi)$. Scaling of medians identifies $b(\xi) = \beta(\xi)/m(\xi)$, as asserted. □

Proof of Theorem 1.1. On each $I_k = [1/k, k]$, $k \geq 2$, construct the family in Lemmas 2.3 and 2.4, and set

$$\widehat{D}_h^{\xi} = b(\xi)^{-1} F_h^{\xi}.$$

Positive scalar multiplication is jointly continuous in the scalar and the lower semicontinuous topology: both the liminf inequality and the recovery condition pass directly through convergent positive factors. Lemma 2.5 therefore gives continuity on I_k. For each deterministic ξ, the modification identity gives $\widehat{D}_h^{\xi} = \beta(\xi)^{-1} D_h^{\xi}$ almost surely.

On any overlap, the two constructions agree almost surely at every point of a fixed countable dense set of parameters. Continuity makes them agree throughout the overlap. Intersecting the resulting countably many probability-one events produces a jointly measurable, continuous metric-valued family on $(0, \infty)$. Assign the Euclidean metric outside this event. All approximations were measurable functions of the original field, so this is the required same-field modification. □

3 Uniform estimates and local comparison

Fix a compact interval $I = [a, b] \Subset (0, \infty)$. Throughout this section, h is a whole-plane GFF with $h_1(0) = 0$, and

$$D^{\theta} = m(\theta)^{-1} D_h^{\theta}, \qquad D = D^{\xi}, \quad E = D^{\eta}, \quad \tau = |\xi - \eta|, \quad q(\theta) = \theta Q(\theta), \quad S_{\theta}(w, r) = r^{q(\theta)} e^{\theta h_r(w)}.$$

Here $m(\theta) = \exp \mathbb{E} \log D_h^{\theta}(0,1)$; its finiteness and compact boundedness follow from Proposition 3.1. Write $A_{r,R}(w) = B_R(w) \setminus \overline{B}_r(w)$ and

$$E_{\rho}(x, y; U) = \inf_{x' \in \overline{B}_{\rho}(x) \cap U,\, y' \in \overline{B}_{\rho}(y) \cap U} E(x', y'; U),$$

omitting U for unrestricted distances. An around-distance is the infimum of lengths of paths whose traces separate the two boundary components of the annulus. All paths below are Euclidean continuous. We call such a separating trace a loop; simplicity is unnecessary.

Probabilistic estimates are uniform in each deterministic parameter or pair in I. An assertion about all spatial points or paths is always on one event for the indicated parameter pair.

3.1 The uniform input

Proposition 3.1 (Compact-uniform estimates)**.** *The function q is positive and Lipschitz on I, and m, m^{-1} are bounded there. There are $p, c, C > 0$ with the following properties.*

1. *For any prescribed finite collection of fixed circular annuli, their normalized across- and around-costs X satisfy, for $t \geq 1$,*

$$\mathbb{P}[X < e^{-t}] \leq C e^{-ct^2}, \qquad \mathbb{P}[X > e^{t}] \leq C \exp\{-ct^2 / \log(e + t)\}. \tag{3.1}$$

In particular, X has positive and negative moments of every fixed order, uniformly on I. At (w, r) the normalization is exactly $S_{\theta}(w, r)$.

2. *If $z \neq w$ are deterministic and U is a connected open set containing them, then*

$$\sup_{\theta\in I} \mathbb{E}D^\theta(z,w;U)^p < \infty, \qquad \sup_{\theta\in I} \mathbb{E}D^\theta(z,w)^{-s} < \infty \quad (s>0). \tag{3.2}$$

For $w \in \partial B_r(z)$, the boundary-internal estimate is

$$\mathbb{E}\left[\left(S_\theta(z,r)^{-1} D^\theta(w,z;\overline{B}_r(z))\right)^p\right] \le C. \tag{3.3}$$

Stopping at $\partial B_{r/2}(z)$ gives the same bound within $\overline{A}_{r/2,r}(z)$. Consequently the probability that more than λr of the outer circle requires normalized access cost greater than T is at most $C\lambda^{-1}T^{-p}$.

3. *For every R and $\chi > \max_{\theta\in I}\{q(\theta)+2\theta\}$, there is $H = H_{\theta,R} \ge 1$ such that*

$$D^\theta(u,v) \ge H^{-1}(|u-v|\wedge 1)^\chi \quad (u,v\in B_R), \qquad \mathbb{P}[H>T] \le C_R T^{-c}. \tag{3.4}$$

There are also $c, K > 0$ such that, for $W \ge 8(R+1)$,

$$\mathbb{P}[\text{some finite } D^\theta\text{-geodesic with endpoints in } B_R \text{ leaves } B_W] \le C(R+1)^K W^{-c}. \tag{3.5}$$

The same moment assertions before normalization hold for D^θ_h.

The proof is in Appendix A. Its main ingredient is a compact-uniform version of the LFPP annular estimates. We prove that version there, rather than assuming locally bounded constants in a fixed-parameter theorem.

Lemma 3.2 (Access at deterministic points)**.** *Fix a deterministic z and $\alpha \in (0,1)$. There are $p, s, C, u_0 > 0$ such that, for $0 < u < u_0$ and $v > 0$, except with probability $Cv^{-p}u^s$, simultaneously for every $r \le u$ there are a loop surrounding $\overline{B}_r(z)$ and a path from z to that loop, contained together in $B_{r^\alpha}(z)$, whose total D^θ-length is at most v. The loop can be chosen in $A_{r,4r}(z)$.*

This is also proved in the appendix. We use it only at finitely many deterministic endpoints. It provides no access assertion for arbitrary points of a supercritical Euclidean ball.

3.2 Gaussian blocks and good annuli

Lemma 3.3 (Gaussian block estimate)**.** *Let $\mathcal{G}(U)$ be the closed Gaussian space generated by smooth zero-mass tests supported in a bounded window U. For windows of diameter at most r at mutual separation at least Ar, the operator correlation of their Gaussian spaces is at most CA^{-2}. For a window of radius r and a fixed-shape annular window of radius $R \ge Cr$ outside it, the bound is Cr/R.*

Suppose a finite collection $U_1, \dots, U_N$ has sum of off-diagonal operator correlations at most one in each row. For $F_j \in \sigma(\mathcal{G}(U_j))$,

$$\mathbb{P}\left[\bigcap_{j=1}^N F_j\right] \le \prod_{j=1}^N \mathbb{P}[F_j]^{1/2}. \tag{3.6}$$

In particular N suitably separated trials, each succeeding with probability at least p, all fail with probability at most $e^{-pN/2}$. For every $B > 0$, a sufficiently high fixed one-trial confidence ensures that fewer than $N/2$ successes has probability at most e^{-BN}.

Proof. The covariance norm of a zero-mass test is a constant multiple of its homogeneous $H^{-1}(\mathbb{R}^2)$ norm. Rescale the two windows and put $k(z) = -\log|z|$. In the separated case replace the covariance kernel by

$$k(Ae + y - x) - k(Ae + y) - k(Ae - x) + k(Ae).$$

The kernel and its derivatives through order one in each variable are $O(A^{-2})$. For nested windows use $k(y-(r/R)x)-k(y)$, with bounds $O(r/R)$. Zero mass leaves the covariance unchanged. Fixed cutoffs and two applications of H^{-1}–H^1 duality give the operator bounds, uniformly in finite-dimensional projections. The row bound implies that the joint covariance is at most twice its block diagonal. The Gaussian product inequality [CDP13, Theorem 1(i)], with all exponents equal to two, proves (3.6). Conditional expectations onto increasing finite Gaussian projections extend it to the closed spaces. Finally, sum (3.6) over subsets of $\lceil N/2 \rceil$ failed trials: the bound is $2^N(1-p)^{N/4}$. □

Local normalized metric events belong to these Gaussian sigma-fields when their pinning circles lie in their observation windows. Circle measures are obtained by approximation in covariance norm. We use well-separated windows along a circle, where the row sums converge like $\sum k^{-2}$, and concentric annuli with a sufficiently large fixed ratio of radii, where they converge geometrically. Each event uses its own pin. Physical ratios between parameters are controlled separately by

$$\left|\log \frac{S_\eta(w,r)}{S_\xi(w,r)}\right| \le C_I\tau|\log r| + \tau|h_r(w)|. \tag{3.7}$$

Corollary 3.4 (Many good annuli)**.** *Fix $\alpha \in (0,1)$, $P > 0$, and $R < \infty$. For each small $u > 0$, outside an event of probability at most $C_R u^P$, every $y \in B_R$ has at least $c\log u^{-1}$ pairwise disjoint annuli surrounding $\overline{B}_{2u}(y)$ and contained in $B_{u^\alpha}(y) \setminus \overline{B}_{2u}(y)$, on each of which*

$$D^\theta(\text{around } A) \le CD^\theta(\text{across } A) < \infty.$$

Proof. Use an $u/10$-grid and radii between $8u$ and $u^\alpha/8$, separated by a sufficiently large fixed ratio. The buffered observation annuli are $A_{r/2,4r}$, and there are $c\log u^{-1}$ of them at each anchor. Choose the one-annulus confidence in Proposition 3.1 so that Lemma 3.3 gives failure Cu^{P+3} at an anchor. A union bound over $O_R(u^{-2})$ anchors proves the result. The buffers give the conclusion for all y using their nearest anchor. □

3.3 Sewing recovery paths

The filled hull of a compact connected planar set is its union with all bounded complementary components. We use the following planar fact repeatedly: filled hulls of two disjoint compact connected sets are either disjoint or nested. Indeed each set lies in one component of the complement of the other; a bounded component gives containment of filled hulls, and otherwise the hulls are disjoint.

Lemma 3.5 (Sewing)**.** *Let $x_0, \dots, x_m \in \mathbb{C}$. For each $1 \le i < m$, suppose a finite-E-length loop $L_i \subset B_{r_i}(x_i)$ surrounds $\overline{B}_{\varepsilon_i}(x_i)$ without meeting it. Suppose a rectifiable path Q_i joins an ε_{i-1}-neighborhood of x_{i-1} to an ε_i-neighborhood of x_i, for $1 \le i \le m$. No loop is required at x_0, x_m. Put $r_+ = \max_{1\le i<m} r_i$ (zero if there are no loops) and $\rho = \max\{\varepsilon_0, \varepsilon_m\}$. There is a target path between the $(\rho + 2r_+)$-neighborhoods of x_0, x_m of length at most*

$$\sum_i \text{len}_E(Q_i) + \sum_i \text{len}_E(L_i). \tag{3.8}$$

The path uses only supplied traces. The conclusion remains valid if the two incident connectors at a vertex use different inner radii and different surrounding loops.

Proof. Keep only maximal filled hulls of the L_i. Distinct retained hulls which intersect have intersecting traces, by the planar fact above. Group retained traces into components of their intersection graph. Within one component, any two trace points can be joined at cost at most the sum of its constituent loop lengths, by following a simple path in that graph. The unions of filled hulls of distinct components are disjoint, and their boundaries lie in their trace networks.

Each interior endpoint ball lies inside one such component. Trim each connector between its last exit from the initial component and first subsequent entry into the terminal one. At an original endpoint keep the actual connector endpoint if it lies outside all retained hulls; otherwise replace it by a point on a retained trace whose hull contains it. This moves it by at most $2r_+$ and costs no length, since the endpoint itself is free to move. The resulting connectors form a walk between the initial and terminal networks or endpoints. Erase cycles, and join successive connectors inside each visited network. Each network is visited once, giving (3.8). For two loops at one vertex, duplicate the vertex and insert a constant connector at its center. Both incident endpoint balls then have their own hull. □

3.4 Short loops and rough comparison

Lemma 3.6 (Two-metric short loops)**.** *Fix a bounded set K, $\alpha \in (0,1)$, and $P > 0$. There are $\beta, c, C > 0$ such that, if $\tau \log \varepsilon^{-1} \le c$, then with probability at least $1 - C\varepsilon^P$ the following holds for every $z \in K$. Any D-geodesic with endpoints outside $B_{\varepsilon^\alpha}(z)$ which visits $B_\varepsilon(z)$ admits a loop $L \subset A_{\varepsilon,\varepsilon^\alpha}(z)$ surrounding $\overline{B}_\varepsilon(z)$ with*

$$\operatorname{len}_E(L) \le C\varepsilon^\beta \operatorname{len}_D(P). \tag{3.9}$$

The right-hand side may instead use any bracketing source excursion. The same assertion holds for an internal source geodesic, provided $B_{2\varepsilon^\alpha}(z)$ is contained in its internal domain. On the same event this holds for the pathwise Weyl metrics $D^\theta_{h+f} := e^{\theta f} \cdot D^\theta$, simultaneously for bounded Lipschitz f satisfying

$$\varepsilon^\alpha \operatorname{Lip}(f) \le 1, \qquad \tau \|f\|_\infty \le 1. \tag{3.10}$$

In coordinates centered at w with scale r, the right-hand side of (3.9) acquires the exact factor $S_\eta(w,r)/S_\xi(w,r)$.

Proof. At an $\varepsilon/10$-grid use separated annuli $A_{r_j,2r_j}$ with $4\varepsilon \le r_j \le \varepsilon^\alpha/8$. Call an annulus good if its source around-cost X_j, source across-cost Y_j, and target around-cost satisfy

$$mS_\xi \le X_j \le MS_\xi, \qquad Y_j \ge mS_\xi, \qquad E(\text{around } A_j) \le MS_\eta.$$

Uniform tails and Lemma 3.3 give $c \log \varepsilon^{-1}$ good annuli at every anchor, except with probability $C\varepsilon^P$. Separately impose $|h_{r_j}(w)| \le A \log \varepsilon^{-1}$ on the $O(\varepsilon^{-2} \log \varepsilon^{-1})$ anchor-radius pairs. Their variances are $O(\log \varepsilon^{-1})$, so a Gaussian union bound has the same failure probability. Equation (3.7) then gives $X_j \le C_0 Y_j$ and $E(\text{around } A_j) \le C_1 X_j$.

Order the good annuli about the nearest anchor from outside to inside, $A_1, \dots, A_J$. A source loop in A_i shortcuts the portion of the geodesic between its first and last intersections. That portion crosses every inner good annulus. Consequently

$$X_i \ge C_0^{-1} \sum_{j>i} X_j, \qquad \sum_{j=1}^{J} X_j \le C_0 \operatorname{len}_D(P).$$

Thus $X_J \le C_0(1 + C_0^{-1})^{-(J-1)} \operatorname{len}_D(P)$, which proves (3.9). Approximate loop infima within a factor two. All events precede the choice of the geodesic. The shortcuts lie in $B_{2\varepsilon^\alpha}(z)$, so the same argument applies to the stated internal geodesics.

Under (3.10), the oscillation of f on each annulus relative to its center is at most two. Weyl scaling changes C_0, C_1 by fixed factors, since the remaining inter-parameter factor is $e^{(\eta-\xi)f(w)} \le e$. The same shortcut proof applies to the shifted geodesics. Exact coordinate scaling gives the stated pin ratio. □

All simultaneous shift statements in this section concern pathwise Weyl weighting. For every prescribed countable deterministic catalogue of shifts these are almost surely the corresponding shifted-field metrics, by fixed-shift Weyl scaling. This is the scope needed for the finite field changes below.

Lemma 3.7 (Loops before first entry)**.** *Fix $\chi \in (0,1)$ and bounded K. There are $p, \beta, c, C > 0$ such that, with probability at least $1 - Cu^p$, the following holds for all $\varepsilon \le u < c$. Let $B_{\varepsilon^\chi}(z) \subset U$, $z \in K$, and let $f \ge 0$ be continuous and vanish outside V. If an internal $e^{-\xi f} \cdot D$-geodesic $P : [0,T] \to U$ has endpoints outside $B_{\varepsilon^\chi}(z)$ and*

$$P(s) \in B_\varepsilon(z), \qquad s \le \inf\{t : P(t) \in V\},$$

where time is shifted-metric length, then

$$D(\text{around } A_{\varepsilon,\varepsilon^\chi}(z)) \le C\varepsilon^\beta s. \tag{3.11}$$

The event works for all the displayed data, including random choices; its constants do not depend on the well's amplitude or regularity.

Proof. Use the same good source annuli as above, with an exponent $\chi' \in (\chi, 1)$. The prefix before s is unchanged by the well, so $\sum_j X_j \le C_0 s$. For a loop in A_i, its first and last intersections bracket s. The prefix from the first intersection to s crosses the inner annuli and has unchanged length. Geodesicity for the shifted metric, together with $e^{-\xi f} \le 1$, gives $X_i \ge C_0^{-1} \sum_{j>i} X_j$. The preceding recurrence proves the bound at one scale. Intersect over dyadic $\varepsilon' \le 2u$. For arbitrary $\varepsilon \le u$, choose $\varepsilon \le \varepsilon' < 2\varepsilon$; then $(\varepsilon')^{\chi'} \le \varepsilon^\chi$ for small u. Summing the failure probabilities proves the assertion. This is the before-entry argument of [DG22, Lemma 2.13], with the constants made uniform by Proposition 3.1. □

Lemma 3.8 (Rough comparison)**.** *For every prescribed $P > 0$ there are constants $c, C, C_* > 0$, depending only on I, P, with the following property. Whenever n is large, $1 \le W \le e^n$, and $\tau n \le c$, outside an event of probability $C(W+1)^2 e^{-Pn}$,*

$$D^\eta_{h+f}(B_{9e^{-n}}(x), B_{9e^{-n}}(y); B_{W+1}) \le C_* \operatorname{len}_{D^\xi_{h+f}}(P) \tag{3.12}$$

for every finite source path $P \subset B_W$ from x to y and every bounded Lipschitz f satisfying $e^{-n}\operatorname{Lip}(f) \le 1$ and $\tau\|f\|_\infty \le 1$. The event can be intersected with its reverse orientation.

Proof. Use a grid of mesh $e^{-2n}/100$ and separated radii in $[4e^{-2n}, e^{-n}/4]$. Uniform tails and the block estimate give, at every anchor, an annulus for which

$$D(\text{across } A_{r/4,r/2}(w)) \ge mS_\xi(w,r), \quad E(\text{around } A_{r,2r}(w)) \le MS_\eta(w,r).$$

The number of pairs is $O((W+1)^2 n e^{4n})$, and $\operatorname{Var} h_r(w) \le \log r^{-1} + 2\log(W+3) + C \le Cn$. Choosing the one-annulus confidence and the pin cutoff An sufficiently large gives the claimed failure for any fixed P. Equation (3.7) and the shift assumptions make the target loop cost at most C_0 times the source crossing cost.

For each point of the source path take the component of its visit to the filled hull of such a loop containing its central visit. A finite inclusion-minimal temporal subcover has multiplicity at most two. Every interval not containing a temporal endpoint crosses its inner annulus and hence pays for its loop at cost at most C_0 times that interval's source length. Their total loop cost is at most $2C_0 \operatorname{len}(P)$. If the two endpoint-clipped intervals overlap, the $9e^{-n}$ endpoint balls intersect. Otherwise discard them and connect the remaining trace networks as in Lemma 3.5. The resulting endpoints move by at most $8e^{-n}$, and all traces lie in B_{W+1}. This proves (3.12). □

The same construction is local in the source-path region. More precisely, if K is a fixed compact subset of an open guard U, then for all sufficiently large n it gives the corresponding comparison for paths in K, with recovery paths in U and failure $C_{K,U}e^{-Pn}$. All observation annuli lie within $O(e^{-n})$ of K. The event is determined by the field in U together with its pinning circle; when that circle is contained in U, it is an event for the field in U modulo constants. This follows by restricting the grid in the proof to a fixed neighborhood of K contained in U.

Lemma 3.9 (Source attachments)**.** *Suppose x, y lie on a D-geodesic and have surrounding D-loops L_x, L_y enclosing their ε-balls. If the full geodesic's endpoints lie outside both filled hulls, then*

$$D(x, y) \le D(B_\varepsilon(x), B_\varepsilon(y)) + 2\operatorname{len}_D(L_x) + 2\operatorname{len}_D(L_y).$$

The assertion is also intrinsic in a guard containing the geodesic segments, recovery paths, and loops in question.

Proof. Geodesicity bounds the segment between first and last hits of L_x by its length. Hence x can be joined to any point of L_x for cost at most twice that length, and similarly for y. Intersecting traces give the conclusion immediately. Disjoint traces have disjoint or nested filled hulls. In the nested case the larger loop shortcuts the whole segment from x to y. In the disjoint case every recovery path crosses both loops; trim it and attach its endpoints as above. □

In particular, for a bounded catalogue well $0 \le f \le M$ with $\operatorname{Lip}(f) \le L$ and source excursions of length at most T_0, Lemmas 3.6 and 3.8 give $D^{\xi}_{h-f}(x, y) \le C_* D^{\eta}_{h-f}(x, y; U) + e$ at buffered source geodesic points. Choose the microscopic recovery radius after M, L, T_0, e: it must make the loop error at most e, satisfy the local oscillation condition, and obey $\tau(|\log \varepsilon| + M) \le c$. The choices below record these restrictions whenever this consequence is used.

3.5 From local seeds to a global comparison

We now fix the local geometry. These choices depend on I, but not on the accuracy used later. Fix the failure exponent in Lemma 3.8 sufficiently large for the uses below and retain its constant C_*. Let $V = A_{1/8,8}$ and choose a fixed high confidence $1 - p_*$. The value of p_* will be decreased once to apply Lemma 3.3 with a prescribed exponential failure rate. Choose M so that the source and target around-costs of $A_{2,3}$ are at most $M/2$. Choose $h_0 > 0$ so that, for each of D, E, every path in $A_{1/2,2}$ of Euclidean diameter at least $1/200$ has length at least h_0. A finite net of small fixed annuli and (3.1) give this last reference event. Also choose H_{pin} so that $\sup_{3/4 \le \rho \le 1} |h_\rho(0)| \le H_{\mathrm{pin}}$, and put

$$K_{\mathrm{pin}} = \exp\{bH_{\mathrm{pin}} + \max_I q \log(4/3)\}.$$

Choose these three references so that their combined failure probability is at most $p_*/10^4$. On this one event, every scale unit with radius in $[3/4, 1]$ is within a factor K_{pin} of the parent unit.

For the finite-forcing argument we need a thin block much cheaper than leaving V, with an exit reference accurate enough for all offset seeds. Choose

$$A_{\mathrm{exit}} > 10^4(C_*+1)K_{\mathrm{pin}}^2 e^{2b}, \qquad m_{\mathrm{exit}} = A_{\mathrm{exit}}M/J, \tag{3.13}$$

where J is a sufficiently large integer. The four source and target crossing costs of $A_{1/4,1/2}$ and $A_{2,3}$ obey

$$\mathbb{P}[\text{one is below } m_{\mathrm{exit}}] \le Ce^{-c(\log J)^2} \le p_*/(10^5 J).$$

Choose J so this holds and $J > 2Me^{2b}/h_0$. The definition of A_{exit} also gives

$$J > 100(C_*+1)K_{\mathrm{pin}}^2 e^{2b}M/m_{\mathrm{exit}}. \tag{3.14}$$

Set $p_0 = p_*/(1000J)$. Thus the exit failure is at most $p_0/100$; the other finite-forcing references can now be chosen at their required fractions of p_0. This joint choice of J, m_{exit} precedes the favorable event and every accuracy-dependent choice.

Put $\alpha = (3/4)^{1/J}$ and $\rho_k = \alpha^k$, $0 \le k < J$. Choose $m > 0$ so that all crossing costs for both metrics of $A_{\alpha^{k+1},\alpha^k}$ are at least m, with combined failure probability at most $p_*/10^4$. Let

$$H_\sigma = A_{\alpha,1} \cap \{\sigma(z) > 0\}, \quad \sigma \in \{\Re, -\Re, \Im, -\Im\}, \qquad T_{\mathrm{seed}} = 2e^{2b}K_{\mathrm{pin}}M/J. \tag{3.15}$$

For $c_0 \in [0, C_*+1]$, an *offset seed* at precision ε is a local event implying

$$E_\varepsilon(P(0), P(T); V) \le c_0 T \tag{3.16}$$

for each intrinsic source geodesic in $\overline{H}_\sigma$ joining its circular sides, of length $T \le T_{\mathrm{seed}}$ and Euclidean diameter at most $1/50$. The event is measurable from the internal metrics in the fixed annular window. The bounded-height all-endpoint inequality $E_\varepsilon(x, y; V) \le c_0 D(x, y; \overline{H}_\sigma)$ is a sufficient version. At an offset $(w, \rho_k r)$ we apply the seed to $h(w + \rho_k r\,\cdot) - h_{\rho_k r}(w)$, so its physical coefficient is exactly

$$c_0 S_\eta(w, \rho_k r)/S_\xi(w, \rho_k r). \tag{3.17}$$

We also use a weaker *grouped seed*. It is one local event such that each intrinsic source geodesic in $\overline{A}_{1/2,2}$, from the inner to the outer boundary and of length at most M, contains a subgeodesic of source length between m and M/J, with endpoints in $A_{3/4,1}$, which satisfies a target comparison at relative precision ε and coefficient c_0. The block is one of the eligible crossings in (3.15). Its bracketing excursion has endpoints at distance at least $1/4$ from both block endpoints. Requiring the offset seed at all J offsets supplies these blocks, with the exact coefficient (3.17): take the shortest of the successive shell crossings. Its diameter is below $1/200$ by the h_0 reference, so after rescaling it fits a half-annulus and has diameter below $1/100$; its own-pin length is at most $K_{\mathrm{pin}}M/J$.

The grouped form permits the favorable argument to choose any shell which works. A positive-probability witness is pigeonholed among shells only after that probability has been obtained. Common reference failures are paid once, whereas using the separate offset seeds costs a union bound over J events.

The exit references also localize the comparisons used in finite forcing. For either metric F and compact endpoint sets in the fixed core $A_{3/5,11/10}$,

$$F(K_1, K_2) \ge \min\{F(K_1, K_2; V), m_{\mathrm{exit}}\}. \tag{3.18}$$

Every path leaving V crosses one of the two exit annuli. Thus an internal lower bound below m_{exit} is global, and a global connector cheaper than m_{exit} stays in V after an arbitrarily small approximation. Equation (3.14) puts all selected thin-block thresholds below this barrier, including their bounded pin distortions. No accuracy-dependent observation window is needed.

Proposition 3.10 (Annular amplification)**.** *There are constants $c, C, K, L > 0$ and, for every $\delta \in (0, 1/2)$, a precision $\varepsilon_\delta \geq c\delta^L$ with the following property. Suppose the offset seed (3.16) at precision ε_δ holds with probability at least $1 - p_*/(100J)$. Alternatively, suppose the grouped seed holds with probability at least $1 - p_*/100$; its common references are included once in this probability. If*

$$n \geq C\delta^{-L-2} + C\delta^{-L}\log(R+1), \qquad \tau n \leq c\delta, \tag{3.19}$$

then except with probability $C(R+1)^K\delta^{-1}e^{-c\delta^L n}$,

$$E_{\exp(-c\delta^L n)}(x, y) \leq (c_0 + \delta)D(x, y) \quad \text{for all } x, y \in B_R \text{ with } D(x, y) < \infty. \tag{3.20}$$

Both orientations may be intersected with a factor-two probability loss. A seed at any finer precision suffices: enlarge its endpoint balls to ε_δ. The depth in (3.19) does not depend on that finer precision.

Proof. We first prove a guarded assertion for every source geodesic in B_W. Its proof uses no access to its original endpoints in the target metric. Put $\theta = m/(2M)$, reducing it to at most one if necessary, and $q_0 = (1+\theta)^{-1}$. Apply Lemma 3.6 with exponent $\alpha = 1/2$ and failure power two on a fixed annular neighborhood of $A_{3/4,1}$. Denote its constants by C_s, β, c_s. Choose

$$\lambda_\delta = (1 - q_0)\delta/32, \qquad u_\delta = \min\{u_0, (m\lambda_\delta/(100e^2C_sM))^{1/\beta}\}, \quad \varepsilon_\delta = u_\delta/10, \tag{3.21}$$

$$N = \max\left\{1, \left\lceil \frac{\log(32(C_*+1)/\delta)}{\log(1+\theta)} \right\rceil\right\}, \quad a_j = 2^{-8}16^{-(j-1)}, \qquad a_\delta = a_N. \tag{3.22}$$

Here u_0 is fixed small enough that the loop event has failure below $p_*/100$ and $u_0^{1/2} < 1/100$. Take

$$n \geq Ca_\delta^{-1}(\log u_\delta^{-1} + \log(N+1) + 1). \tag{3.23}$$

Both u_δ and a_δ are bounded below by fixed powers of δ, and $N \leq C\log(2/\delta)$. After enlarging L, these choices satisfy the polynomial bounds in the statement.

At each site the composite local event comprises the fixed references, the offset seeds (or the grouped seed), and the underlying local event of Lemma 3.6 at radius u_δ. All are determined by the field modulo constants in $A_{1/32,16}$ in site coordinates. The loop event is local because its construction uses only annuli around the possible block endpoints; the geodesics occur only in its conclusion. Its hypothesis $\tau\log u_\delta^{-1} \leq c_s$ follows from (3.19). The composite event has probability at least $1 - p_*$. In particular, its fixed-relative loop failure is paid inside the local event, before amplification over sites.

In band j take radii in $[e^{-4a_jn}, e^{-a_jn}]$, with sufficiently large fixed ratio R_0, and a spatial grid of mesh $e^{-4a_jn}/1000$ in $B_{W+1/10}$. There are $O((W+1)^2e^{8a_jn})$ anchors and at least ca_jn radii at each. Choose p_* so Lemma 3.3 beats this grid entropy. Except with probability $C(W+1)^2e^{-c_1a_jn}$, at least half the radii are composite-good at each anchor, where c_1 can be fixed larger than ten. Impose separately $|h_r(w)| \leq An$ at all these radii and offsets. There are $O(Nn(W+1)^2e^{8a_1n})$ such pairs and their variances are $O(n + \log(W+1))$, so a Gaussian union bound costs at most CNe^{-2n} for fixed large A. For $1 \leq W \leq e^{a_\delta n/100}$, all bands and pins therefore hold except with probability

$$CNe^{-ca_\delta n}. \tag{3.24}$$

On this event, (3.7) makes every local coefficient at most $b_\delta = c_0 + \delta/8$, after decreasing c in (3.19). A short target loop at a block endpoint has length at most

$$eC_su_\delta^\beta MS_\xi(w, r), \tag{3.25}$$

since the surrounding source excursion has length at most $MS_\xi(w,r)$. This calculation uses the site's scale and pin in both metrics.

Here is the deterministic improvement in one band. Assume for every subgeodesic of $P:[0,\ell]\to B_W$ that

$$E_{\rho_{j-1}}(P(s),P(t);B_{W+1})\le C_{j-1}(t-s),\quad \rho_0=9e^{-n/2},\quad \rho_j=20e^{-a_jn}.$$

At every point of P choose a good site containing it in $B_{r/10}(w)$, and use a source loop in $A_{2r,3r}(w)$ as a primary filled-hull domain. A finite minimal cover by the corresponding temporal components has multiplicity at most two. There is at most one component clipped at each endpoint. If these overlap, the ρ_j endpoint balls intersect and there is nothing to prove. Otherwise let their total length be $l\ell$.

Every unclipped component has length at most $MS_\xi(w,r)$ by a primary-loop shortcut. After its central visit, restrict to the passage from the last hit of $\partial B_{r/2}(w)$ before the first subsequent hit of $\partial B_{2r}(w)$. This is an intrinsic geodesic in the annular window. The grouped seed supplies a thin block of length at least $mS_\xi(w,r)$. Two-color the interval cover and retain the color whose blocks have greater total length T. Its blocks have disjoint interiors and

$$T\ge\theta(1-l)\ell. \tag{3.26}$$

Their endpoints are at distance at least r from the primary loop. Thus the local short-loop event applies to the whole primary excursion, enclosing balls of radius $u_\delta r$ with loops of outer radius $u_\delta^{1/2}r$. Equation (3.25) and the lower block length bound make the sum of their endpoint-loop lengths at most

$$(2eC_su_\delta^\beta M/m)\ell\le\lambda_\delta\ell/4.$$

Both local and old recovery endpoints lie inside these loops. The local radius is at most $u_\delta r/10$; the old radius satisfies

$$\rho_{j-1}<u_\delta e^{-4a_jn}/100. \tag{3.27}$$

For $j\ge2$, this follows from $a_{j-1}=16a_j$ and $12a_jn>\log(2000/u_\delta)$; for $j=1$ use $n/2-4a_1n>n/3$. Both follow from (3.23). Hence the bands are processed from smaller physical scales to larger ones.

Use improved connectors on the selected blocks and old connectors in the gaps. Lemma 3.5, allowing distinct loops at a shared endpoint, gives target cost at most

$$[C_{j-1}-\theta(C_{j-1}-b_\delta)(1-l)+\lambda_\delta]\ell.$$

Its endpoint displacement is at most $\rho_{j-1}+2u_\delta^{1/2}e^{-a_jn}<3e^{-a_jn}$. All local connectors lie in their fixed annular guards, hence in B_{W+1}. Alternatively discard the two clipped components and apply the old comparison to the remaining subgeodesic. Its cost is $C_{j-1}(1-l)\ell$ and the new endpoint displacement is at most $6e^{-a_jn}+\rho_{j-1}<\rho_j$.

For $C\ge b\ge0$, the minimum of $C-\theta(C-b)(1-l)$ and $C(1-l)$ is at most $b+(C-b)/(1+\theta)$: maximize the minimum over $l\in[0,1]$ by equating its increasing and decreasing affine branches. Retain an already better coefficient if $C_{j-1}<b_\delta$. We obtain

$$(C_j-b_\delta)_+\le q_0(C_{j-1}-b_\delta)_++\lambda_\delta. \tag{3.28}$$

There is no additive old error, so the number of blocks causes no loss and no absolute lower bound for their lengths is needed.

Initialize with Lemma 3.8 at logarithmic scale $n/2$, choosing its failure exponent once so its failure is at most $C(W+1)^2e^{-2n}$. Iterating (3.28) gives

$$C_N\le b_\delta+q_0^N(C_*+1)+\lambda_\delta/(1-q_0)\le c_0+\delta.$$

All events were chosen before the source geodesic. This proves the guarded conclusion, simultaneously for every such geodesic, with recovery $20e^{-a_\delta n}$ and failure (3.24).

Finally take $W = e^{a_\delta n/100}$. Proposition 3.1 confines every finite source geodesic between points of B_R to B_W, except with probability $C(R+1)^K W^{-c}$. Increasing the depth by $Ca_\delta^{-1}\log(R+1)$ ensures $W \geq 8(R+1)$. Fixed-parameter geodesic existence applies to every finite $D(x,y)$. Since $a_\delta \geq c\delta^L$, the recovery radius can be enlarged to $e^{-c\delta^L n}$, and $N \leq C\delta^{-1}$ gives the stated failure probability. □

4 A local comparison with high probability

Fix $I = [a,b] \Subset (0,\infty)$. We write

$$D = m(\xi)^{-1}D_h^\xi, \qquad E = m(\eta)^{-1}D_h^\eta, \qquad \tau = |\xi - \eta|, \qquad q(\theta) = \theta Q(\theta).$$

Constants in this section depend only on I and fixed Euclidean domains, unless stated otherwise. The unit of the metric with parameter θ at (z,r) is $S_\theta(z,r) = r^{q(\theta)}e^{\theta h_r(z)}$. The normalizing factor $m(\theta)$ is already included in the unit-scale metric. In particular, when $|h_r(z)| \leq Cn$ and $|\log r| \leq Cn$, the ratio of the two units is between $e^{-C\tau n}$ and $e^{C\tau n}$. Write $C_b = C_*$ for the fixed rough-comparison constant chosen before (3.13).

We use Proposition 3.10 first with a conditional local input, and later with the local input proved below. The construction of the well follows [DG22, Sections 4–5]. Two features are useful here: its target endpoints need only be regular for the target metric, and all the comparisons in the multiplicity bound will be made at one source field.

4.1 A favorable target arc

The fixed annuli in the next lemma can be translated, scaled, and rotated. We retain the parent circle-average pin when selecting a thin shell inside an annulus.

Lemma 4.1 (Favorable arcs)**.** *There are fixed $\alpha \in (3/4,1)$, an integer J, and positive constants $m_0, M_0, m_{\mathrm{exit}}$ with the following property. For $0 < v < v_0$, there is a deterministic list of at most Cv^{-C} labels. A label specifies one of the J shells with outer radius in $[3/4,1]$, a coordinate half-annulus H in that shell, two rounded gate centers u_0, v_0 on its circular sides, and $s \in [t,\sqrt{t}] \cap \{4^{-k} : k \in \mathbb{N}\}$, where $t \geq cv^C$ and $s \leq (1-\alpha)/200$. At least one label has probability at least cv^C of the following event, measurable in the fixed annulus $V_0 = A_{1/8,8}(0)$ modulo the unit pin.*

There is an arc $\Gamma \subset \overline{H}$ which is a subarc of an intrinsic E-geodesic across the buffer $\overline{A}_{1/2,2}$, with endpoints $u \in B_{s/2}(u_0)$, $v' \in B_{s/2}(v_0)$ on its inner and outer circular sides, respectively, and length W, such that

$$m_0 \leq W \leq m_{\mathrm{exit}}/4, \qquad W \leq (1+Cv)D(G_u,G_v), \qquad G_u = B_{2s}(u_0), \quad G_v = B_{2s}(v_0). \tag{4.1}$$

The half-annulus cutline has distance at least a fixed positive constant from the arc and its gates. The access estimate

$$\Big|\{x \in \partial B_s(u_0) : E(u,x;\overline{B}_s(u_0)) > \lambda W\}\Big| \leq \lambda s/2, \qquad \lambda = cv, \tag{4.2}$$

holds, and so does its counterpart at v'. The event also includes source exit barriers of height m_{exit} in $A_{1/4,1/2}$ and $A_{2,3}$. All these conclusions hold whenever

$$n_{\mathrm{fav}}(v) = \lceil Cv^{-K}\rceil, \qquad \tau \leq cv^{K+1}, \tag{4.3}$$

for fixed $C, K, c > 0$.

Proof. Logarithmic normalization and Proposition 3.1 give $\mathbb{E}\log E(0,1) = \mathbb{E}\log D(0,1) = 0$ and a uniform second moment for both logarithms. Cantelli's inequality therefore gives

$$\mathbb{P}[E(0,1) \le (1+v)D(0,1)] \ge cv^2. \tag{4.4}$$

Use the fixed J, α and exit references of (3.13)–(3.15), chosen for both metric orientations; write M_0 for their fixed loop bound. For an eligible intrinsic target excursion across $A_{r/2,2r}(z)$ of length at most $M_0 S_\eta$, in each of the J radial bands take the segment from the last inner-boundary visit before its first outer-boundary visit. These intervals are disjoint. One has length at most $M_0 S_\eta / J$. The fixed diameter reference in that choice bounds its diameter by $r/200$, and the fixed shell crossing references bound its length below by $m_0 S_\eta$. One signed coordinate stays larger than half the shell radius along this segment, so it lies in one of the four half-annuli with the stated cutline clearance.

Recall why these choices are compatible: with $m_{\mathrm{exit}} = A_{\mathrm{exit}} M_0 / J$ and fixed large A_{exit}, the lower logarithmic Gaussian tail makes exit failure $O(e^{-c(\log J)^2}) = o(J^{-1})$. A large fixed J gives both the required confidence and the short-crossing cap. All these choices precede v.

For a radius r_{buf} specified below, call such an arc favorable if, in parent unit coordinates,

$$W \le (1+4v) D(B_{r_{\mathrm{buf}}}(u), B_{r_{\mathrm{buf}}}(v'); V_0). \tag{4.5}$$

This implies the same inequality with global distance. Indeed, both endpoint balls lie in $A_{0.6,1.1}$; a path leaving V_0 crosses one of the two source barriers and costs at least $m_{\mathrm{exit}} \ge 4W$. A path staying in V_0 costs at least the right-hand denominator in (4.5).

Put $p_f = p_*/400$. The fixed common references above have total failure at most $p_*/400$. The union of the favorable events over the fixed shells has probability at least p_f. Otherwise, outside an event of probability at most $p_*/200$, every eligible target excursion has a source connector between the buffered endpoint balls of length at most $W/(1+4v)$. This is precisely the grouped reverse local input to Proposition 3.10. With coefficient allowance a small multiple of v and $\tau n \le cv$, it gives a source path between exponentially small neighborhoods of $0, 1$ of length at most $E(0,1)/(1+3v)$. The uniform point moments and geodesic confinement allow the required height and spatial guard to be chosen as fixed powers of v^{-1}, with total exception smaller than $cv^2/8$. Choose a fixed A such that $\mathbb{P}[D(0,1) < v^A] \le cv^2/8$. Lemma 3.2 attaches the fixed endpoints at total cost v^{A+2}, with another exception smaller than $cv^2/8$. On an event of positive probability intersecting (4.4), we would then have

$$D(0,1) \le \frac{1+v}{1+3v} D(0,1) + v^{A+2}, \qquad D(0,1) \ge v^A,$$

which is impossible for small v. For the quantitative cutoff, both amplification and attachment errors are bounded by a fixed power of v^{-1} times $\exp(-cv^L n)$ for a fixed L. Taking the fixed K in (4.3) larger than $L+1$ and the amplifier's minimum-depth exponent makes these errors smaller than their allocated multiples of v^2, and supplies the required guard. The same choice gives $\tau n_{\mathrm{fav}} \le cv$.

It remains to choose gates. Around an endpoint of the short arc, the larger intrinsic target excursion has both endpoints a fixed distance away. The local proof of Lemma 3.6, with the two metrics equal and all its loops inside the buffer, bounds target escape to radius $2\sqrt{t}$ by $Ct^{\beta/2} M_0$, for a fixed $\beta > 0$. Its all-point exceptional probability is less than $p_f/4$ once the fixed power in the choice of t is large enough. For a deterministic center and radius s, impose a target loop upper bound and target crossing lower bound with constants R and R^{-1} in $A_{s/2,s}$. Also require that all but circle arclength $\lambda s/2$ of ∂B_s can reach $\partial B_{s/2}$ inside the closed annulus $\overline{A}_{s/2,s}$ at cost at most RS_η. The closed-disk stopped-access bound (3.3), integrated over the circle, gives failure at most CR^{-p}/λ for the last requirement. Thus $R \le C\lambda^{-C}$ makes the combined one-annulus failure any desired fixed constant.

Round each circular boundary at mesh $\lambda t/100$. Lemma 3.3, applied to the scales in $[t, \sqrt{t}]$ and then to a union over $O((\lambda t)^{-1})$ centers, shows that, outside an event of probability less than $p_f/4$, each center is good at more than three quarters of these scales. The two rounded endpoint centers share a good scale. Their rounding error is less than $t/50$, so the true endpoints lie in the half-radius gate balls. The target escape path crosses the good annulus. Its lower crossing bound makes the local target unit at most $CRt^{\beta/2}M_0$. A boundary access path, the good separating loop, and the endpoint escape path can consequently be joined at cost at most $CR^2t^{\beta/2}M_0$ inside the closed gate disk.

Fix $\beta_s > 0$ so that the permitted local precision in Proposition 3.10 obeys $\varepsilon_\delta \geq c\delta^{1/\beta_s}$. Choose t to be a sufficiently small fixed power of λ, so that

$$CR^2t^{\beta/2}M_0 \leq \lambda^2 m_0, \qquad t \leq \lambda(1-\alpha)^2, \qquad K_{\mathrm{buf}}\sqrt{t} \leq cv^{1/\beta_s}. \tag{4.6}$$

This also makes the preceding union-bound error small and ensures $s \leq (1-\alpha)/200$. Set $r_{\mathrm{buf}} = K_{\mathrm{buf}}\sqrt{t}$, with the fixed K_{buf} large enough that both enlarged gates $B_{2s}(u_0), B_{2s}(v_0)$ lie in the corresponding buffered balls. The last inequality is the recovery-radius hypothesis used in the reverse amplification above. The first proves (4.2). There are only polynomially many choices of shell, half-annulus, rounded centers, and scale. Pigeonholing the event of probability at least $p_f/2$ proves the lemma. □

4.2 Attaching a target endpoint to a source geodesic

For a nonnegative smooth function $f \leq M$, write $\widehat{D} = e^{-\xi f} \cdot D$ and $\widehat{E} = e^{-\eta f} \cdot E$. The following deterministic lemma is the only place where the two-metric well argument differs substantially from its one-parameter counterpart.

Lemma 4.2 (Gate attachment)**.** *Suppose $f = M$ on $\overline{B}_R(z_0)$ and is supported on V, where $B_{R+\omega}(z_0) \subset V$ and $\omega \ll d \ll R$. Let P be a $\widehat{D}$-geodesic, parametrized by length. Assume it has a central segment as in (B.4), with every exterior component of diameter at most d, and has outer bracketing times $a' < a < b < b'$ on $\partial B_{R+\omega+100d}(z_0)$. Its length on $[a', b']$ is at most A, where $A \geq 1$. Suppose $u \in B_{R/2}(z_0)$ and*

$$\begin{aligned} &|\{x \in \partial B_R : E(u, x; \overline{B}_R) > \lambda W\}| \leq \lambda R/2, \\ &|\{x \in \partial B_R : \mathrm{dist}(x, P) \leq 2\omega\}| \leq \lambda R, \qquad |P \cap \partial B_{R+\omega}| = 0. \end{aligned} \tag{4.7}$$

For some $H \geq 1$, at every relevant center and every $\omega \leq r \leq 100d$, assume

$$E(\text{around } A_{2r,3r}) \leq r^{-1/4} D(\text{across } A_{r/4,r/2}), \qquad D(\text{across } A_{r/4,r/2}) \geq H^{-1} r^{\gamma}. \tag{4.8}$$

Assume also, for points x, y on the source segment, that

$$\widehat{D}(x, y) \leq C_b \widehat{E}(x, y; \Omega) + e_*. \tag{4.9}$$

Here Ω contains all the loops and gate paths under discussion. If

$$d \leq \left[\frac{R}{CC_b(1 + \log(AH) + \gamma)}\right]^{16}, \qquad e_* \leq \lambda e^{-\eta M} W, \tag{4.10}$$

then some $t \in (a', b')$ satisfies

$$P(t) \in B_{2R}(z_0), \qquad \widehat{E}(u, P(t); \Omega) \leq C\lambda e^{-\eta M} W. \tag{4.11}$$

Proof. If u lies on the central segment there is nothing to prove. Otherwise apply Lemma B.3 to this segment and $B_{R+\omega}(z_0)$. For x on this circle, let x' be its radial projection to $\partial B_R(z_0)$. The radial lifts of the two exceptional sets in (4.7) have lengths at most λR and $2\lambda R$.

If the interior blocked arcs have union of length greater than $3\lambda R$, choose a point of this union outside both exceptions. A target path from u to x' of length at most $2\lambda W$, followed by the radial segment to x, must hit the blocking crosscut. The radial segment avoids P, so the target path itself hits P. It lies in the plateau disk and hence acquires the full factor $e^{-\eta M}$, proving (4.11).

In the other case, (B.5) and $|P(a) - P(b)| \geq R/8$ give exterior arcs of total length at least $R/8 - 3\lambda R$. Keep their maximal members, whose interiors are disjoint. Arcs of length at most ω have total length at most $2\lambda R$, since their radial projections are within 2ω of an endpoint on P. Arcs containing no point with good target access have total length at most λR. The remaining arcs I therefore satisfy, for small fixed λ,

$$\omega \leq |I| \leq 2d, \qquad \sum_I |I| \geq R/16, \tag{4.12}$$

and each contains a point with access cost at most λW. Its genuine exterior bracket lies outside V, by the central segment condition. This assertion concerns only that central segment.

Write $\ell = |I|$ and let y_I be the first endpoint of I hit by P. Choose a target separating loop $L_I \subset A_{2\ell,3\ell}(y_I)$ of length at most twice the upper bound in (4.8). Let s_I be the first subsequent hit of $\partial B_{\ell/4}(y_I)$, and t_I the first subsequent hit of L_I. These times occur before b', since

$$L_I \subset B_{R+\omega+6d}(z_0), \qquad |P(b') - y_I| \geq 100d > 3\ell.$$

The initial crossing from radius $\ell/4$ to $\ell/2$ occurs on the exterior bracket: its endpoint chord is at least $(2/\pi)\ell > \ell/2$. On this crossing the well vanishes. Until its first loop hit the path stays in the filled hull of L_I, so

$$P([s_I, t_I]) \subset B_{3\ell}(y_I), \qquad \Delta_I = t_I - s_I \geq H^{-1}\ell^{\gamma}. \tag{4.13}$$

A good target access path from u to x' intersects L_I, because $|x' - y_I| \leq \ell + \omega \leq 2\ell$ whereas $|u - y_I| > 3\ell$. The access path receives the plateau discount and the loop can only become shorter under the well. Hence

$$\widehat{E}(u, P(t_I); \Omega) \leq 2\lambda e^{-\eta M} W + 2\ell^{-1/4}\Delta_I. \tag{4.14}$$

The mass bound in (4.12) implies that for some $k \geq \lfloor \log_2(1/(Cd)) \rfloor$, at least $cR2^{k/2}$ arcs have lengths in $[2^{-k-1}, 2^{-k})$. Otherwise the sum of their lengths would be at most $CR2^{-k_0/2} < R/16$. A bounded-overlap selection leaves this many, up to a fixed factor, whose balls $B_{3|I|}(y_I)$ are disjoint. Their time intervals in (4.13) are disjoint. The lower bound there and the upper bound A put their lengths into at most $C(k + B)$ dyadic bins, where $B = 1 + \log(AH) + \gamma$ is enlarged to be positive. One bin contains N chronologically ordered intervals, of lengths in $[q_*, 2q_*]$, with

$$N \geq \frac{cR2^{k/2}}{k + B}.$$

For their final points $x_1, \ldots, x_N$, the target triangle inequality, (4.9), and (4.14) give

$$(N - 1)q_* \leq CC_b(\lambda e^{-\eta M} W + 2^{k/4} q_*) + e_*.$$

The first condition in (4.10) ensures $CC_b(k + B)2^{-k/4}/R$ is sufficiently small for every such k. Absorb the q_* term to obtain $q_* \leq CC_b\lambda e^{-\eta M} W/N$. Substituting this in (4.14) proves the result. The final points are within $6d$ of the gate circle and hence belong to $B_{2R}(z_0)$. □

4.3 A well which forces an exact shortcut

Write $T_0(x) = x$ and $T_{j+1}(x) = \exp(T_j(x))$. The tower heights below are fixed; their values will be immaterial.

Proposition 4.3 (The one-site well)**.** *For $0 < v, p < 10^{-3}$, put $X = 10 + (vp)^{-1}$. There is a finite deterministic catalogue of triples (U, V, f) and local events $\mathcal{A}_f$, at least one of which has probability at least $1 - p$, with the following properties. Each triple satisfies*

$$U \Subset V \Subset A_{3/2,5/2}, \qquad 0 \le f \le M, \qquad f = M \text{ on } \overline{U}, \qquad \operatorname{supp} f \subset V.$$

On $\mathcal{A}_f$ there are source barriers of height a_0 between V and $\partial A_{1,3}$, an unchanged separating source loop $\Lambda \subset A_{3,13/4}$ of length at most A_0, and a separating source loop $\Pi \subset U$ of length at most L_0. We take $a_0 \le 1 \le A_0$ and

$$b_0 = \frac{a_0^2}{4A_0}, \qquad e^{-aM} L_0 \le a_0/4, \qquad a_0, b_0, s_0 \ge X^{-C}, \quad A_0, L_0 \le X^C. \tag{4.15}$$

Let P be any global $\widehat{D} = e^{-\xi f} \cdot D$-geodesic with both endpoints outside B_4. If it visits V, let τ_0, σ_0 be its first and last visits to $\overline{V}$ surrounding its visits to V. Then its entire intervening portion lies in $A_{1,3}$. In particular all visits to V belong to one component of $P^{-1}(B_4)$. The interval from the last hit of Λ before τ_0 to the first hit after σ_0 is contained in B_4 and contains all visits to V. If

$$D(P(\tau_0), P(\sigma_0); B_4) \ge b_0, \tag{4.16}$$

there are $\tau_0 \le s < t \le \sigma_0$ such that

$$t - s \ge s_0, \qquad \widehat{E}(P(s), P(t); A_{1,4}) \le (1 + Cv)(t - s). \tag{4.17}$$

Times are measured in $\widehat{D}$-length.

The events are determined by the field modulo its unit pin in $A_{1/2,7/2}$, and the catalogue consists of at most X^C canonical geometries. Its functions satisfy

$$M \le C \log X, \qquad \operatorname{Lip} f + \|f\|_\nabla^2 \le T_2(X^C). \tag{4.18}$$

The assertion holds under a parameter cutoff $\tau_{\text{site}}(v, p) \ge T_2(X^C)^{-1}$. Every function vanishes near the unit circle, so all these changes preserve the unit pin exactly.

Proof. We choose the geometric scales once, in the order described below. Every choice through the well height is a fixed power of X; the collar and the final auxiliary scale are the only smaller scales. Allocate a fixed fraction of p to each of the finitely many probabilistic requirements.

The annular moments choose $a_0 \ge X^{-C}$ and $A_0 \le X^C$ for the two barrier annuli $A_{1,3/2}, A_{5/2,3}$ and the loop in $A_{3,13/4}$. Lemma 3.7 chooses $a_3 \ge X^{-C}$ and an exponent $\theta > 0$ for source loops around first-entry and last-exit points. Choose $a_4 \ge X^{-C}$ with

$$a_4 \le a_3/10, \qquad C A_0 a_4^\theta \le b_0/10. \tag{4.19}$$

Choose a favorable label from Lemma 4.1 and place its copies at scale ρ around ∂B_2, as in Lemma B.1. Its favorable probability is $p_{\text{fav}} \ge cv^C$. Every circle arc of length $a_4/4$ contains at least $ca_4\lambda/\rho$ chamber centers. Their fixed inspection annuli are separated by a large fixed multiple of their diameters. Lemma 3.3 gives

$$\mathbb{P}[\text{no favorable chamber on such an arc}] \le \exp\{-cp_{\text{fav}} a_4 \lambda/\rho\}.$$

Choose $\rho \geq X^{-C}$ small enough that

$$\rho \ll \lambda a_4, \qquad A_0\rho^\theta \leq \lambda a_0, \qquad p_{\mathrm{fav}} a_4 \lambda/\rho \geq C\log(C/(pa_4)). \tag{4.20}$$

A union bound over C/a_4 deterministic arcs gives a favorable chamber in every arc of length $a_4/2$. Impose separately $|h_\rho(z) - h_1(0)| \leq C\log X$ at all chamber centers. This has the required probability by the Gaussian circle-average bound. The ratio of the two physical units is then $e^{O(\tau \log X)}$. Thus the favorable denominator remains valid with coefficient $1 + Cv$ whenever $\tau \log X \leq cv$.

The annular moment estimates and a dyadic spatial net give, with the required probability, constants $H \leq X^C$ and fixed γ such that

$$E(\text{around } A_{2r,3r}(y)) \leq r^{-1/4} D(\text{across } A_{r/4,r/2}(y)), \quad D(\text{across } A_{r/4,r/2}(y)) \geq H^{-1} r^\gamma \tag{4.21}$$

for all relevant centers and all $0 < r \leq 100d$, once $d \geq X^{-C}$ is a sufficiently small fixed power of X^{-1}. Here the net uses fixed buffered subannuli, so the same estimates hold at arbitrary centers and intermediate radii by containment. To check the cross-metric estimate at every small scale, the Gaussian pin bound gives a unit ratio at most $r^{-C\tau}$; the annular moments make the remaining ratio at most $r^{-1/8}$ with summable failure after the spatial union. A fixed small upper bound on τ absorbs the pin ratio into $r^{-1/4}$. The same net gives the source spatial lower bound in the guard. In particular this step makes no restriction involving the collar width, which has not yet been chosen.

Decrease d so that $d \ll \lambda(1-\alpha)t\rho$ and

$$d \leq \left[\frac{t\rho}{CC_b(1 + \log(A_0 H) + \gamma)}\right]^{16}. \tag{4.22}$$

Next choose $a_6 \geq X^{-C}$ with

$$a_6 \ll \min\{a_3, d, \lambda t\rho(1-\alpha)\}, \qquad CA_0(2a_6)^\theta \leq \lambda H^{-1} d^\gamma. \tag{4.23}$$

Lemma B.1, with $a = a_6$, gives a grid cycle of clearance $10\delta_0$, with $\delta_0 \geq X^{-C}$, within $a_6/4$ of every outer boundary point of U. Place source loops in $A_{\delta_0,2\delta_0}$ at its grid vertices. Adjacent loops intersect: their enclosed δ_0-balls overlap, and neither filled hull can contain the other's enclosed ball, since $3\delta_0/2 + \delta_0 > 2\delta_0$. Joining consecutive intersections along the loops gives a separating trace $\Pi \subset U$. Each joining piece is homotopic to its grid edge in a small ball avoiding the central hole, so the trace has winding number one. It remains within a_6 of every outer boundary point. There are at most X^C loops, and their positive moments give a total length $L_0 \leq X^C$ with the required probability. There is one canonical family for each favorable label, so a union over all families costs only a polynomial factor.

Choose the well height

$$M = \frac{1}{a}\log\left(\frac{100 L_0 H}{\lambda \min\{a_0, d^\gamma\}}\right), \tag{4.24}$$

increased by a fixed constant if necessary. Then $M \leq C\log X$, $e^{-aM} L_0 \leq a_0/4$, and $e^{-aM} L_0 \leq \lambda H^{-1} d^\gamma$.

We next make the boundary collar thin. For a fixed annulus, there is probability at least $ce^{-C(M+1)^2}$ that its outer source loop costs at most $\frac{1}{2}e^{-2bM}$ times its inner source crossing. Indeed, start with fixed positive-probability annular upper and lower references, and subtract a smooth function of height $(2b/a)M + C_I$ on the outer subannulus, vanishing on the inner subannulus. Choose C_I so that its Weyl factor, at most $e^{-a[(2b/a)M + C_I]}$, gives the required ratio uniformly for $\xi \in [a, b]$. The Cameron–Martin Cauchy–Schwarz bound costs at most $e^{C(M+1)^2}$. Such an annulus cannot be

crossed by a geodesic which visits its central ball, for any pathwise Weyl change by a function of absolute value at most M: its outer loop is cheaper than the bracketed inner crossing. Applying Lemma 3.3 at the separated scales between ε and $\sqrt{\varepsilon}/4$ therefore bounds the probability of coming within ε of a fixed boundary point by $C\varepsilon^{c\exp[-C(M+1)^2]}$. Integration along a deterministic curve of length T, followed by Markov's inequality, bounds the probability of approaching more than L units of its arclength by

$$C(T/L)\varepsilon^{c\exp[-C(M+1)^2]}. \tag{4.25}$$

The same event works for all such changes and all the relevant geodesics, because every near-curve set is contained in the single field-determined set having no favorable annulus.

Apply this to the plateau boundaries and all gate circles, whose total length and number are at most X^C. We can choose $\omega \ll \min\{a_6, d\}$ with

$$\log \omega^{-1} \asymp e^{C(M+1)^2} \log X \tag{4.26}$$

so that every relevant changed source geodesic approaches these curves within 2ω on total arclength at most $\lambda t\rho$. Applying (4.25) at decreasing radii also gives zero arclength intersection with every enlarged gate circle $\partial B_{s\rho+\omega}$, almost surely. Let $V = B_\omega(U)$ and let f be M times a smooth mollification of the indicator of $B_{\omega/3}(\overline{U})$, using a mollifier of radius less than $\omega/3$. Then

$$\operatorname{Lip} f \le CM/\omega, \qquad \|f\|_\nabla^2 \le CM^2/\omega^2.$$

These prove (4.18).

Only now choose an auxiliary microscopic radius ε. All target paths used for the gate comparison lie in $A_{5/4,13/4}$; use its buffered rough comparison, whose auxiliary annuli lie in $A_{1/2,7/2}$. The before-entry events likewise need annuli only near the entry points in $A_{3/2,5/2}$. Lemmas 3.8, 3.6, and 3.9 give (4.9), simultaneously on source geodesic portions of length at most $2A_0$, with error at most $C\varepsilon^\beta X^C$ and failure at most $C\varepsilon^{p_1}$, for fixed positive exponents. The physical favorable arc lengths are at least X^{-C} on the chamber pin event. It is enough to choose

$$\varepsilon^{1/2} CM/\omega \le 1, \qquad C\varepsilon^\beta X^C \le \lambda e^{-bM} X^{-C}, \qquad C\varepsilon^{p_1} \le p/20. \tag{4.27}$$

A fixed power of ω times a fixed power of X^{-1} suffices. The parameter requirements are

$$\tau(\log \varepsilon^{-1} + M + \log X) \le cv \quad \text{and} \quad \tau \le cv^{K+1}. \tag{4.28}$$

After increasing C, the reciprocal $T_2(X^C)^{-1}$ satisfies both, including the favorable cutoff (4.3).

We prove the deterministic assertions on the resulting event. First P cannot enter B_1. Such a visit would be bracketed by two hits of Π; geodesicity bounds the intervening output length by $e^{-\xi M} L_0 < a_0$, whereas the unchanged crossing of $A_{1,3/2}$ costs at least a_0. The unchanged outer loop brackets all visits to V and bounds the output length between its relevant hits by A_0.

Consider the first component of $P^{-1}(B_4)$ which visits V. Restrict its geodesic to start at the last outer-loop hit before τ_0. Its prefix before τ_0 is unchanged by the well, has length at most A_0, and stays in $A_{1,4}$. Apply Lemma 3.7 there, and do the time-reversed construction in the last component which visits V. This gives source loops around $P(\tau_0)$ and $P(\sigma_0)$, in $A_{2a_6,\sqrt{2a_6}}$, of lengths at most $CA_0(2a_6)^\theta$. The two points are within ω of the outer boundary of U: their exterior path tails avoid V. Both little loops intersect the dense plateau trace Π. Following these three traces gives a connector between a point of P before τ_0 and a point after σ_0. Global geodesicity now proves

$$\sigma_0 - \tau_0 \le CA_0(2a_6)^\theta + e^{-\xi M} L_0 \le 3\lambda H^{-1} d^\gamma. \tag{4.29}$$

No confinement of the middle portion was used to obtain this bound. Its first upper bound is less than a_0 by (4.20), $a_6 \ll \rho$, and (4.24). The unchanged barriers force that portion to remain in $A_{1,3}$. Each of its components outside V has diameter at most d, since otherwise an initial crossing inside the source guard would cost at least $cH^{-1}d^{\gamma}$, contradicting (4.29). The tails from the last preceding outer-loop hit and the first subsequent hit stay in that loop's filled hull. Their union with the middle portion is therefore contained in B_4. This also proves that the enlarged intervals for sites with disjoint B_4 inspections are disjoint in time.

Under (4.16), the trigger points have Euclidean separation at least a_4. If they were closer, the before-entry estimate at scale a_4 would give a source loop surrounding both, of length at most $CA_0a_4^{\theta}$. The last hit before entry and first hit after exit bracket the two points. Geodesicity bounds the output length of that bracket by the loop length; its two exterior tails have the same source length. The tails and the loop would give a source connector inside B_4 of length less than b_0, by (4.19).

Lemma B.2 now supplies a favorable chamber and the two central gate segments. At each gate use the outer radius $s\rho + \omega + 100d$. These balls and all target loops lie inside the enlarged favorable gates and $A_{1,4}$, since $\omega + 100d \ll s\rho$. The small arcs have length at most $2d$, covered by the range $100d$ in (4.21). The target access comes from (4.2); the source avoidance comes from (4.25); and (4.22), (4.27) give precisely the two conditions in (4.10). Lemma 4.2 attaches each favorable endpoint to P at target cost at most $C\lambda e^{-\eta M}W$. Concatenating with the favorable arc gives

$$\widehat{E}(P(s), P(t); A_{1,4}) \le (1 + C\lambda)e^{-\eta M}W.$$

The global favorable denominator in (4.1) and $f \le M$ give $t - s \ge e^{-\xi M}W/(1+Cv)$. Thus (4.28) and $\lambda = cv$ prove the coefficient in (4.17). The two attachment points lie in distinct enlarged gates with separation much greater than d. The source spatial lower bound gives $t - s \ge ce^{-bM}H^{-1}d^{\gamma} \ge X^{-C}$. This proves the proposition. □

4.4 Finite field changes

We first record the two deterministic reductions needed for counting. They use neither a finite diameter for Euclidean balls nor target access to arbitrary endpoints.

For $r > 0$, take an $r/16$-net of the bounded endpoint region, and at each net point a set $K_a = \overline{B}_{r/8}(a)$ and $L_a = \overline{B}_{r/2}(a)$. If $E_r(x, y) > cD(x, y)$ for $c \ge 0$, choose net points within $r/16$ of x, y. Then

$$E(L_a, L_{a'}) > cD(K_a, K_{a'}), \qquad D(K_a, K_{a'}) \le D(x, y). \tag{4.30}$$

Indeed $L_a \subset B_r(x)$ and $L_{a'} \subset B_r(y)$. If $|x - y| \ge d_0$ and $r \le d_0/4$, the compact sets are separated by at least $d_0/2$. There are $O(r^{-4})$ ordered pairs. Lower semicontinuity and compactness give a minimizing endpoint pair whenever its set distance is finite, and hence a set geodesic. Fix one measurable canonical choice for every field and endpoint pair. For example, the deterministic-compact-set uniqueness statement of [DG22, Lemma 2.7], together with absolute continuity for our finite deterministic catalogues, gives this choice simultaneously at every occurrence of the same output field.

Here is a convenient consequence of the first-to-last assertion in Proposition 4.3.

Lemma 4.4 (Deleting failed triggers)**.** *Suppose k sites have disjoint full B_4 inspections, disjoint from the entire endpoint sets. Their source units lie in $[Q, 2Q]$, their source references from Proposition 4.3 hold, and the source set geodesic visits every central unit disk. There is a subset of at least κk sites, where*

$$\kappa = \frac{a_0}{8A_0}, \tag{4.31}$$

such that the geodesic after subtracting exactly their wells satisfies the source trigger at every retained site. For these sites, intervals containing all support visits can be restored by unchanged source collar loops, at cost at most $2A_0Q$ per site, and the restoration intervals are disjoint in time.

Proof. On the original source geodesic, replace the portion bracketing each central-disk visit by its plateau loop. The disjoint inspections make these temporal replacements disjoint. Each original portion costs at least a_0Q, whereas its discounted loop costs at most $e^{-aM}L_0$ times that site's unit. Using the pointwise bound $e^{-aM}L_0 \le a_0/4$ in that same unit gives a saving of at least $3a_0Q/4$ per site. If T_Z is the distance after inserting all k wells and $T_\varnothing$ the unmodified distance, then

$$T_\varnothing - T_Z \ge 3a_0Qk/4.$$

Delete one failing site at a time. If the current output geodesic never visits its collar, deletion does not increase its length. Otherwise Proposition 4.3 puts all its collar visits in one B_4 component. The trigger fails for the first and last visits of that component, so there is an intrinsic source connector inside B_4 of cost less than b_0 times the site unit. Replace that entire intervening segment by this connector when deleting the well. All other wells vanish on this inspection. Thus deletion increases the distance by at most $2b_0Q \le a_0Q/2$. The local references at every remaining site are unchanged, so the argument applies again after each deletion. At termination, the distance is still at least $a_0Qk/4$ below $T_\varnothing$.

For the remaining m sites, enlarge each first-to-last collar interval to its adjacent unchanged outer-loop hits. Proposition 4.3 puts each enlarged interval inside its full inspection, so these intervals are time-disjoint. Restore them by source outer-loop subpaths, each of cost at most $2A_0Q$. Outside their union the output and source lengths coincide. This gives $T_\varnothing - T_{Z'} \le 2A_0Qm$, so $m \ge a_0k/(8A_0)$. The use of the local proposition with other wells present is valid pathwise: at the site in question the input field and all its local metrics are unchanged by every other well. □

We now improve a comparison coefficient by a deterministic amount. An offset seed below always means the bounded-height local event (3.16), with the fixed geometry (3.15). We will in fact obtain its stronger form simultaneous over all pairs on the two circular sides, with intrinsic source distance at most T_{seed}. The source and target unit pins are the same.

Proposition 4.5 (Improving a coefficient)**.** *There are fixed $q_0, p_0 > 0$, a fixed R_s, and fixed tower height d_1 such that the following holds. For each $0 < v < v_0$ there are $g, r_*, \varepsilon_*, \tau_* > 0$ satisfying*

$$T_{d_1}(v^{-C})^{-1} \le g, r_*, \varepsilon_*, \tau_* \le v. \tag{4.32}$$

Suppose $\tau \le \tau_*$ and $1 + C_0v \le c \le C_b$, where C_0 is a fixed sufficiently large constant and C_b is the fixed rough comparison constant. Then

$$\mathbb{P}[E_{\varepsilon_*}(x,y) \le cD(x,y) \text{ for all } x, y \in B_{R_s}] \ge 1 - q_0 \tag{4.33}$$

implies that the offset seed with coefficient $c - g$ and precision r_* has probability at least $1 - p_0$. Here $p_0 = p_*/(1000J)$ as fixed above and $q_0 < p_0/100$. The same scales admit a reset by Proposition 3.10: the resulting seed implies (4.33) with $c - g/2$ in place of c.

Proof. Apply Proposition 4.3 at a fixed sufficiently high confidence, to be specified for the spatial covering below. Denote its shortcut coefficient by $b_* \le 1 + C_1v$, its minimum shortcut length by s_0, its label count by J_v, and the largest Dirichlet norm and Lipschitz norm of a unit-scale well by M_f, L_f. Together with κ, a_0, A_0, M, all these quantities or their reciprocals, as appropriate, are bounded by a fixed tower in v^{-1}. In fact $s_0, \kappa \ge v^C$, $A_0 \le v^{-C}$, and $M \le C \log v^{-1}$.

Put $\Delta = e^{-\ell}$. The local whole-site event has a fixed observation window, independent of its confidence. At each trial radius r, predetermine the candidate centers on $(r/100)\mathbb{Z}^2$. Use a separate $\Delta^2/100$-net only to index spatial coverage; an anchor's trial at radius r uses its nearest predetermined $r/100$-lattice center. The displacement is at most $r/100$, so the buffered windows at sufficiently separated trial radii remain separated, and the central ball contains the anchor's nearby points. The block estimate and the fine spatial net show, outside an event of probability $C\Delta^P$, that every point of a fixed source guard belongs to an eligible central ball at some geometric radius in $[\Delta^2, \Delta]$. Here fix $P > 10$ first, and then choose the one-site confidence sufficiently high: at each of $O(\Delta^{-4})$ anchors, failure through $c\ell$ separated annuli is exponentially small with an arbitrarily large fixed rate. A separate Gaussian pin cap gives, at every candidate center and radius, source units between Δ^{K_0} and Δ^{-K_0} for a fixed K_0.

For a violating thin block, the target exit barrier in (3.18) turns its internal violation into a global one, since $(c-g)T_{\mathrm{seed}} < m_{\mathrm{exit}}$. Its endpoints have a fixed separation $d_0 \geq 3(1-\alpha)/4$, and its global source distance is at most $U_0 = T_{\mathrm{seed}}$. Choose $r_* = \Delta^A$ with fixed A to be specified. Equation (4.30) gives at most $C\Delta^{-4A}$ deterministic endpoint pairs K_1, K_2, with larger balls L_1, L_2, such that

$$E(L_1, L_2) > (c-g)D(K_1, K_2), \qquad D(K_1, K_2) \leq U_0, \qquad \mathrm{dist}(K_1, K_2) \geq d_0/2. \tag{4.34}$$

Order the block radially. The center z_1 of its first endpoint ball then has modulus at most $\alpha + O(r_*)$. Set $r_c = \min\{1-\alpha, d_0\}/32$. Before the source set geodesic first leaves $B_{4r_c}(z_1)$, take its first visits to radial levels between $2r_c$ and $3r_c$, spaced by a large multiple of Δ. The eligible central balls at these visits give $c\Delta^{-1}$ disjoint candidate full inspections, all in $B_{5r_c}(z_1) \Subset B_1$ and a fixed distance from both entire endpoint sets.

Pigeonhole by radius, by one of finitely many grid colors making full inspections disjoint, and by a dyadic bin of the source unit. There are at most $C\ell^C$ resulting deterministic catalogues. One contains

$$N = \lceil \Delta^{-1/2} \rceil \tag{4.35}$$

eligible sites with units in one bin $[Q, 2Q]$, where $Q \geq \Delta^{K_0}$. Each catalogue has at most $C\Delta^{-D_0}$ physical sites, for a fixed D_0. On the pin cap, the physical target/source unit ratio is $e^{O(\tau\ell)}$. The restriction (4.45) below makes this $1 + O(v)$, which is included in the coefficient b_*. Restrict every catalogue, including unselected sites, to the fixed core $B_{5r_c}(z_1)$ and exclude every full inspection meeting an endpoint set. All its shifts are supported strictly inside the original pinning circle. Each well also vanishes on its own site circle, and all other wells vanish on that site's full inspection. Thus every pin and every physical source unit used in the count is exactly preserved.

Choose a fixed outer source barrier, beyond B_4, with crossing cost greater than $2U_0$ and failure smaller than a prescribed fraction of p_0. Proposition 3.1 and scaling give such a deterministic radius: if $\mu = \min_I q > 0$ and p is a small fixed positive number, its inverse moment at radius R is at most $CR^{-\mu p + b^2 p^2} \leq CR^{-\mu p/2}$. Take R_s larger than this radius by a fixed buffer. The source set geodesic is a competitor of length at most U_0 for every negative catalogue shift, so every corresponding output geodesic stays in B_{R_s}. The barrier is unchanged under all catalogue shifts.

Fix an endpoint pair and a catalogue $\mathcal{C}$. Let $B_{\mathcal{C}}$ be the intersection of the event inside (4.33), (4.34), the common regularity references, and the supply (4.35) in this catalogue. For an integer k, put $m_k = \lceil \kappa k \rceil$. For every labelled subset Z with $m_k \leq \#Z \leq k$, let $A_Z(H)$ assert $B_{\mathcal{C}}(H)$, the validity of all selected site labels and units, and the trigger at every selected site for the canonical set geodesic of $H - f_Z$, where f_Z is their sum. On $B_{\mathcal{C}}$ choose the first N eligible sites and their first valid labels. Lemma 4.4 maps every k-subset to an admissible surviving subset. Each survivor has

at most N^{k-m_k} preimages, so

$$\sum_Z \mathbf{1}_{A_Z}(H) \geq N^{m_k} k^{-k} \mathbf{1}_{B_\mathcal{C}}(H). \tag{4.36}$$

For the upper count fix one output field h. All indices with $A_Z(h+f_Z)$ use the same output set geodesic P of length $T = D_h(K_1, K_2) \leq U_0$. Choose one active index Z_0 and write $H_0 = h + f_{Z_0}$, $D_0 = D_{H_0}$, $E_0 = E_{H_0}$. Discard its at most k sites. At each of the remaining m active physical sites, the local proposition supplies an exact E_h shortcut of coefficient b_* and source length $L_i \geq s_0 Q$. Its path lies in that inspection, so has the same E_0 length. All these shortcut intervals are disjoint. For the Z_0 sites, restore every support visit using the first-to-last assertion of Proposition 4.3 and its unchanged collar loops, at source cost at most $2A_0Q$ each. These intervals are also disjoint from the shortcuts.

At every interior junction use Lemma 3.6, imposed at the one source H_0 simultaneously for smooth shifts, with shift $-f_{Z_0}$. It produces target loops for h; transfer them to H_0 at cost at most e^{bM}. Disjoint supports make the height of the sum at most M, and its Lipschitz constant at most $L_f \Delta^{-2}$. The full endpoint sets have a fixed buffer from all junctions. Choose the microscopic scale so that each transferred loop has length at most u_* and outer radius at most $r_*/16$. On residual source pieces outside the selected supports, the D_0 and output lengths coincide. Apply the comparison at H_0 to these pieces and to the source restoration connectors, then use the exact shortcuts at the other sites. Lemma 3.5 joins their recovery paths, using only interior junction loops. Its final endpoints belong to L_1, L_2. We obtain

$$E_0(L_1, L_2) \leq cT - (c - b_*) \sum_{i=1}^{m} L_i + 2cA_0 kQ + 2(m+k)u_*. \tag{4.37}$$

The source near-bad inequality and nonnegative Weyl monotonicity give $E_0(L_1, L_2) > (c - g)D_0(K_1, K_2) \geq (c - g)T$. Choose C_0 so that $c - b_* \geq c_I v$ for a fixed $c_I > 0$. If

$$gU_0 \leq c_I v s_0 \Delta^{K_0}/100, \qquad u_* \leq c_I v s_0 \Delta^{K_0}/100, \tag{4.38}$$

then (4.37) implies $m \leq K_1(v)k$, where $K_1(v) \leq C(1 + A_0/(vs_0))$. Counting subsets and labels among these physical sites gives

$$\sum_Z \mathbf{1}_{A_Z}(h + f_Z) \leq (CK_1(v)J_v)^k. \tag{4.39}$$

All comparison inequalities in this argument belong to H_0.

We specify the remaining scales. Use Lemma 3.6 with outer-radius exponent $1/2$ and a sufficiently large fixed failure power, and denote its length exponent by β. Set

$$g = \frac{c_I v s_0}{100 U_0} \Delta^{K_0}, \qquad r_* = \Delta^A, \qquad \varepsilon_* = \Delta^B, \tag{4.40}$$

where fixed $A > K_0/\beta_s$ and then fixed B satisfy $B/2 > A + 2$ and $B\beta > K_0 + 2$. Increasing ℓ makes

$$\varepsilon_*^{1/2} L_f \Delta^{-2} \leq 1, \quad (2\varepsilon_*)^{1/2} \leq r_*/16, \quad C e^{bM} \varepsilon_*^{\beta} U_0 \leq c_I v s_0 \Delta^{K_0}/100. \tag{4.41}$$

Thus the preceding sewing is valid; its common-source failure is at most $C\Delta^P$. The same increase makes $r_* \leq c g^{1/\beta_s}$, so this finer seed precision is admissible for amplification at allowance $g/2$.

For completeness, the Cameron–Martin overlap bound is as follows. If shifts f_i of norm at most L satisfy $\sum_i 1_{A_i}(h) \geq L_- 1_B(h)$ and $\sum_i 1_{A_i}(h+f_i) \leq L_+$, with at most N_+ indices, then

$$\mathbb{P}(B) \leq \frac{L_+}{L_-} \min\{N_+, \exp(L\sqrt{2\log N_+})\}. \tag{4.42}$$

Indeed change variables by each deterministic shift and bound by the largest corresponding Gaussian density. Its pth moment gives an expectation at most $N_+^{1/p} e^{(p-1)L^2/2}$; take $p = 1 + \sqrt{2\log N_+}/L$. The alternative bound follows by summing the densities.

Here $N_+ \leq (CJ_v\Delta^{-D_0})^k$. The disjoint supports give $\|f_Z\|_{\nabla}^2 \leq kM_f^2$. Equations (4.36), (4.39), and (4.42) therefore yield

$$\log \mathbb{P}(B_{\mathcal{C}}) \leq -\tfrac{1}{2}\kappa k\ell + k\log k + k\log(CK_1J_v) + kM_f\sqrt{2(D_0\ell + \log(CJ_v))}. \tag{4.43}$$

Choose $k \geq C(4A+P+10)/\kappa$, and then choose ℓ so the last three terms are at most $\kappa k\ell/4$. A fixed polynomial lower bound in k, M_f, κ^{-1} and the logarithms of the other prefactors suffices, as well as ensuring all preceding requirements and $k \leq N$. Consequently $\mathbb{P}(B_{\mathcal{C}}) \leq \Delta^{4A+P+5}$. Sum over the $C\Delta^{-4A}$ endpoint pairs and $C\ell^C$ catalogues. Pay the source comparison exception q_0 and the common references only once. Their total can be made less than $p_0/2$, and the count smaller than $p_0/2$. This proves the claimed local improvement.

To reset confidence, choose a fixed M_{amp} larger than all minimum-depth and error exponents in Proposition 3.10, and set

$$n_{\mathrm{amp}} = \left\lceil Cg^{-M_{\mathrm{amp}}}(B\ell + \log g^{-1} + \log q_0^{-1} + 1)\right\rceil. \tag{4.44}$$

At allowance $g/2$ this includes the minimum-depth condition, makes the final recovery radius at most ε_*, and makes failure at most q_0 on B_{R_s}. The common parameter restriction is

$$\tau \leq \tau_* := c\min\left\{v, \tau_{\mathrm{site}}(v,p), \frac{g}{1+\ell+M}, \frac{v}{1+B\ell+M}, \frac{g}{n_{\mathrm{amp}}}\right\}. \tag{4.45}$$

In particular $\tau n_{\mathrm{amp}} \leq cg$, as required by the reset amplifier. Apply the seed at each of the fixed offsets; their total failure is at most $Jp_0 < p_*/100$. This gives (4.33) with $c - g/2$.

Finally, the chosen ℓ is bounded by a fixed tower in v^{-1}. After ensuring $\log(100U_0/(c_I v s_0)) \leq \ell$, we have $\log g^{-1} \leq (K_0+1)\ell$ and $n_{\mathrm{amp}} \leq e^{C\ell}$. Thus $g/n_{\mathrm{amp}} \geq e^{-C'\ell}$. The one-site cutoff already has a reciprocal fixed-tower bound. Equations (4.40) and (4.45) prove (4.32), after increasing the fixed height. □

4.5 Resetting confidence

Proposition 4.6 (The high-probability local seed)**.** *For the fixed $p_0 = p_*/(1000J)$ there are fixed d, C such that, for $0 < v < v_0$, the offset seed (3.16) with coefficient $1+v$ holds with probability at least $1 - p_0$ whenever*

$$\tau \leq T_d(v^{-C})^{-1}. \tag{4.46}$$

Its precision $r(v)$ satisfies $T_d(v^{-C})^{-1} \leq r(v) \leq cv^{1/\beta_s}$, and the upper bound can be replaced by any prescribed fixed power of v. The assertion holds in both metric orientations, simultaneously for all pairs on the two circular sides whose intrinsic source distance is at most T_{seed}.

Proof. Start (4.33) at $c = C_b$. To justify its global form, use the all-geodesic confinement estimate (3.5): except with probability less than $q_0/4$, every finite source geodesic with endpoints in the fixed B_{R_s} stays in a deterministic larger disk B_W. Apply Lemma 3.8 on B_W at recovery radius ε_* and error probability less than $q_0/4$. It compares the actual finite global geodesic for every endpoint pair. Infinite source distances cause no restriction. The parameter condition and depth needed here follow by the same increase of ℓ and decrease of the fixed constant in (4.45).

Proposition 4.5 gives a local seed at coefficient $c - g$; its reset gives the global comparison at $c - g/2$ with the same confidence $1 - q_0$. Repeat while $c > 1 + C_0 v$. There are at most $2C_b/g$ steps. These are implications between probability bounds, so their exceptional events are never intersected. At each step the confidence returns to $1-q_0$, using the same $\varepsilon_*, r_*, g, n_{\mathrm{amp}}$ and parameter cutoff. The last local seed already has coefficient at most $1+C_0 v$. Replace v by a sufficiently small fixed multiple of the desired accuracy and repeat in the reverse orientation. The bound $r_* \le c g^{1/\beta_s} \le c v^{1/\beta_s}$ and (4.32) give the stated precision and cutoff. Increasing the fixed A and then the lower bound for ℓ in (4.40) makes $r(v)$ smaller than any prescribed fixed power of v, without changing the nature of the tower bound. □

5 Proof of the comparison estimate

Proof of Proposition 1.2. Fix $I \Subset (0, \infty)$, let L_0 be the exponent in Proposition 3.10, and set $v = \delta/8$. Proposition 4.6 gives the bounded-height offset seeds, in both orientations, with coefficient at most $1 + v$ and failure at most $p_0 = p_*/(1000J)$. Use its adjustable precision bound with fixed power $2L_0 + 2$. Since the amplifier permits precision $\varepsilon_{\delta/2} \ge c(\delta/2)^{L_0}$, the seed precision is at most $\varepsilon_{\delta/2}$ after decreasing the fixed upper bound for δ. Enlarge the seed endpoint balls to this precision. This does not change the coefficient or the probability of the seed. In particular, the depth required by the final amplification depends polynomially on δ^{-1}; it does not depend on the smaller precision used in the finite counting argument.

The parameter cutoff in Proposition 4.6 is bounded below by $T_{d_0}(v^{-C_1})^{-1}$ for fixed d_0, C_1. Increasing the fixed height d and exponent C if necessary, the first inequality in (1.4) implies this cutoff. Apply Proposition 3.10 with allowance $\delta/2$. It gives coefficient

$$1 + v + \delta/2 \le 1 + \delta$$

and both reciprocal comparisons on B_R.

Choose $L > L_0 + 2$. The bound $n \ge C\delta^{-L}$ supplies its polynomial minimum depth. If $R \le \exp(c\delta^L n)$ and c is sufficiently small, then

$$C_2 \delta^{-L_0} \log(R+1) \le C_2 \delta^{-L_0} \log 2 + C_2 c \delta^{L-L_0} n \le n/2.$$

Increasing C pays the first term and the remaining minimum-depth requirement. Decreasing c also supplies the restriction $\tau n \le c_2 \delta/2$ of the amplifier. Finally, decrease c and increase C, L once more so that its recovery radius is at most $\exp(-c\delta^L n)$ and the sum of its two failure probabilities is at most (1.5). Enlarging the recovery balls proves (1.6). □

A Uniform annular estimates

This appendix proves Proposition 3.1 and Lemma 3.2. We use the white-noise fields $\Phi_{m,n}$, their finite-range truncations $\Psi_{m,n}$, and corresponding LFPP metrics $D^{\xi}_{m,n}, \widetilde{D}^{\xi}_{m,n}$ of [DG21, Section 2].

Let $L_n^\xi, \widetilde{L}_n^\xi$ be unit-square crossing lengths and write

$$\ell_n^\xi(p) = p\text{-quantile of } L_n^\xi, \quad \lambda_n(\xi) = \ell_n^\xi(1/2), \quad \Lambda_n^\xi(p) = \max_{j\le n} \frac{\ell_j^\xi(1-p)}{\ell_j^\xi(p)}.$$

Constants below are uniform over $\xi \in I = [a,b]$ and the cutoff n. They may depend on a specified finite list of annular or rectangular shapes.

A.1 The LFPP estimate

Proposition A.1. *There exists $p \in (0,1/2)$ such that*

$$\sup_{\xi\in I, n\ge 0} \Lambda_n^\xi(p) < \infty. \tag{A.1}$$

For each fixed circular or square annulus A, $t \ge 1$,

$$\mathbb{P}[D_{0,n}^\xi(\text{across } A) < e^{-t}\lambda_n(\xi)] \le Ce^{-ct^2}, \tag{A.2}$$

$$\mathbb{P}[D_{0,n}^\xi(\text{around } A) > e^{t}\lambda_n(\xi)] \le C\exp\{-ct^2/\log(e+t)\}. \tag{A.3}$$

The same holds for $\widetilde{D}$.

Proof. We give the uniformity argument and a shortened variance induction. For fixed ξ, [DG21, Proposition 2.5] gives $n^{-1}\log\lambda_n(\xi) \to (q(\xi)-1)\log 2$, with $Q(\xi) > 0$. Put $S_n = \sup_{[0,1]^2}|\Phi_{0,n}|$. The integrands along every path give

$$|\log L_n^\xi - \log L_n^\eta| \le |\xi-\eta| S_n, \qquad \mathbb{E}S_n \le Cn,$$

where the second estimate is [DDDF20, Proposition 2, (2.10)]. Gaussian concentration gives

$$\operatorname{Var}\log L_n^\xi \le Cb^2 n, \qquad |\mathbb{E}\log L_n^\xi - \log\lambda_n(\xi)| \le Cb\sqrt{n}.$$

Indeed the pointwise field variance is $n\log 2$, and a field change g changes the log-crossing by at most $\xi\|g\|_\infty$; the mean–median bound follows from Chebyshev. Hence $n^{-1}\mathbb{E}\log L_n^\xi$ are equi-Lipschitz. Their pointwise limit is Lipschitz, and convergence is uniform on I, by a finite-net argument. Thus q is Lipschitz and, for every $\zeta > 0$,

$$C_\zeta^{-1} 2^{(q(\xi)-1-\zeta)n} \le \lambda_n(\xi) \le C_\zeta 2^{(q(\xi)-1+\zeta)n}. \tag{A.4}$$

The finitely many small n are controlled by $|\log L_n^\xi| \le bS_n$. The same concentration argument gives, for fixed p,

$$\Lambda_n^\xi(p) \le e^{C_{b,p}\sqrt{n}}. \tag{A.5}$$

Uniform preliminary tails. Before proving tightness, the rectangle arguments of [DDDF20, Sections 3–4], as used in [DG21, Proposition 2.4], give for a common sufficiently small p and every required fixed-shape crossing or loop observable X_n,

$$\begin{aligned} \mathbb{P}[X_n < e^{-t}\ell_n(p)] &\le Ce^{-ct^2}, \\ \mathbb{P}[X_n > e^{t}\Lambda_n(p)\ell_n(p)] &\le Ce^{-ct^2/\log(e+t)}, \qquad t \ge 1. \end{aligned} \tag{A.6}$$

We record the dependence on ξ, since a fixed-parameter statement alone would not suffice here.

The conformal-map couplings in [DDDF20, Lemma 6] split their field error into a low-frequency supremum with Gaussian tails and an independent high-frequency field with bounded pointwise variance. Their constants depend on the finitely many maps in Lemmas 11–12, not on ξ. Multiplication by $\xi \le b$ keeps them uniform. In Lemma 9 the moment parameter is at least one once $\log \varepsilon^{-1} \ge 2b^2\sigma_F^2$, where σ_F^2 bounds the high-frequency variance. One common small ε therefore works, and both errors cost at most $\exp(C_b\sqrt{\log \varepsilon^{-1}})$ in the quantile comparisons. Derivative bounds, subdivisions, and marked arcs are geometric; positive association (Lemma 13) has no parameter-dependent constant. Thus Proposition 14 and the RSW bounds (3.36)–(3.39) have common constants at all required quantile levels.

The truncation comparison [DDDF20, Proposition 5], equivalently [DG21, Lemma 2.3], supplies a parameter-independent Z_U with

$$\sup_n \|\Phi_{0,n} - \Psi_{0,n}\|_U \le Z_U, \qquad \mathbb{P}[Z_U > s] \le C_U e^{-c_U s^2}. \tag{A.7}$$

Lengths change by factors in $[e^{-bZ_U}, e^{bZ_U}]$. The quantile comparison used here is

$$e^{-R} q_Y(p-\varepsilon) \le q_X(p) \le e^R q_Y(p+\varepsilon)$$

when $\mathbb{P}[|\log X - \log Y| > R] \le \varepsilon$. This fixes the directions needed when passing between the original and truncated fields.

For the lower tail, the separated-rectangle recursion in [DDDF20, Proposition 16] uses only finite-range independence and the preceding RSW constants. With one C_b and sufficiently small p_0 it reads

$$p_{j+1} = (C_b p_j)^2, \qquad r_{j+1} \ge r_j C_b^{-1} e^{-C_b\sqrt{|\log(C_b p_j)|}}.$$

Thus $p_j = C_b^{-2}(C_b^2 p_0)^{2^j}$ while $r_j \ge r_0 e^{-C_b j - C_b 2^{j/2}}$. Taking $2^{j/2}$ of order t gives the first bound in (A.6); the final RSW and truncation comparisons in Corollary 17 use one fixed square quantile.

For the upper tail, the four steps of [DDDF20, Proposition 18] remain uniform as follows. The finite-dependence range and neighbor count in its high-density percolation argument depend only on the truncated kernel. The common high-quantile RSW estimate therefore gives the crossing bound $Ck^2\ell_n(1-p)$ with failure Ce^{-ck} on a $(3k,k)$ rectangle; returning to the original field costs $e^{Cb\sqrt{k}}$. In the coarse/fine decomposition all field estimates are parameter-independent, and the coarse metric factor is bounded using $\xi \le b$. The scale-quantile comparison follows from a union bound over $O(4^k)$ rectangles, the lower tail, and the coarse maximum bound. It gives

$$\ell_n^\xi(p) \ge 2^{-2\xi k} e^{-C\sqrt{k}} \ell_{n-k}^\xi(p).$$

Substitution gives threshold $e^{Cs\sqrt{\log(e+s)}+Cs}\Lambda_n(p)\ell_n(p)$ with failure Ce^{-cs^2} for $1 \le s \le 2^{n/2}$. Beyond this range the straight-line moment estimate [DDDF20, (3.40)] uses only pointwise variance $\xi^2 n \log 2 \le b^2 n \log 2$ and a lower quantile bound from the field infimum. Conversion of $e^{\xi s}$ to e^t uses $a \le \xi \le b$. This proves the second bound. Choose p below the finitely many required thresholds before the argument; every resulting high quantile is bounded by $\ell_n(1-p)$. No monotonicity of $\Lambda_n(p)\ell_n(p)$ in p is needed.

Finite rectangle gluing gives the required annuli. For their dilates by $R \ge 1$, the same covering argument as [DG21, Lemma 3.14] gives

$$\begin{aligned} \mathbb{P}[X_{n,R} < e^{-t}\ell_n(p)] &\le CRe^{-ct^2}, \\ \mathbb{P}[Y_{n,R} > e^t R\Lambda_n(p)\ell_n(p)] &\le CRe^{-ct^2/\log(e+t)}. \end{aligned} \tag{A.8}$$

The lower bound covers a separating curve by $O(R)$ unit annuli; the upper bound uses $O(R)$ overlapping annuli along a connecting curve. Splitting according to $bZ_U \le t/2$ transfers these bounds to the truncated metric, retaining the original quantiles.

A mesoscopic estimate. Fix $\varepsilon_0 \in (0, 1/100)$ in the truncation kernel, and suppose $n > K$, $m = n - K$, and

$$\Lambda_m^\xi(p) \le e^{A\sqrt{K}}. \tag{A.9}$$

Use the dyadic squares $\mathcal{S}_K$ and surrounding annuli of [DG21, Definition 3.5, (3.23)–(3.24)]. Their radii are comparable to $2^{-K}K^{\varepsilon_0}$ and their number is $O(4^K)$. Include the two square-ring shapes used in the crossing construction [DG21, (3.70)–(3.73)]. Put $H_S = \Phi_{0,K}(v_S)$, and let A_S, C_S denote full truncated around- and across-costs. Uniformly for $s \ge C_A\sqrt{K}$, except with probability

$$C\exp\{CK - cs^2/[K^{2\varepsilon_0}\log(e+s)]\}, \tag{A.10}$$

all required annuli satisfy

$$C_S \ge 2^{-K}\lambda_m e^{\xi H_S - s}, \qquad A_S \le 2^{-K}\lambda_m e^{\xi H_S + s}. \tag{A.11}$$

The same holds for the coarse metric, with $\lambda_m = 1$. Indeed the fine-field annuli rescale to dilates of size $O(K^{\varepsilon_0})$. Apply (A.8) and (A.9), then union over $\mathcal{S}_K$. The coarse oscillation estimate follows from [DDDF20, Proposition 3, (2.16)]:

$$\mathbb{P}[2^{-K}\|\nabla\Phi_{0,K}\|_U > x] \le C4^K e^{-cx^2}.$$

An annulus has diameter $O(2^{-K}K^{\varepsilon_0})$, giving the $K^{2\varepsilon_0}$ in (A.10). Finally apply (A.7).

Set $\beta = 2/3$ and $\nu = 2\beta - 2\varepsilon_0 > 1$. For $M_K = \max_S A_S/C_S$, cancellation of the common factors in (A.11) and tail integration give

$$\mathbb{E}M_K^s \le e^{C_s K^\beta} \qquad (s > 0). \tag{A.12}$$

In fact beyond $x = CK^\beta$ the tail of $(\log M_K)_+$ is bounded by (A.10) with s replaced by cx. The negative exponent dominates both $2CK$ and any fixed multiple of x, since $2\beta - 2\varepsilon_0 > 1$ and $\beta > 2\varepsilon_0$.

The deterministic square-chain construction in [DG21, Lemma 3.19] gives $e^{-Cs}\lambda_m\widetilde{L}_K \le \widetilde{L}_n \le e^{Cs}\lambda_m\widetilde{L}_K$ on the simultaneous full and coarse events. To see both directions, record a crossing's successive exits from neighboring dyadic squares. The corresponding disjoint time intervals pay their full-metric ring crossings, while loops in the coarse square rings contain a crossing. The common weights $2^{-K}e^{\xi H_S}$ cancel. Interchange the two metrics for the reverse inequality. Equations (A.4), (A.5), and (A.6), with the $e^{C\sqrt{K}}$ loss absorbed in $2^{\zeta K}$, now give an event $\mathcal{E}$ with

$$\mathbb{P}[\mathcal{E}^c] \le Ce^{-cK^\nu/\log K} \tag{A.13}$$

on which (A.11) holds with $s = CK^\beta$ and

$$2^{(q-1-\zeta)K}\lambda_m e^{-CK^\beta} \le \widetilde{L}_n \le 2^{(q-1+\zeta)K}\lambda_m e^{CK^\beta}. \tag{A.14}$$

Variance contraction. Let $\mathcal{G} = \sigma(\Psi_{K,n})$, $L = \widetilde{L}_n$, and let F_S mean that a chosen crossing geodesic visits the inner disk of the circular annulus A_S. The resampling and Efron–Stein inequalities [DG21, Lemmas 3.6 and 3.8] give, with $Z_S = \mathbb{E}[\mathbf{1}_{F_S}A_S/L \mid \mathcal{G}]$,

$$\mathrm{Var}(\mathbb{E}[\log L \mid \mathcal{G}]) \le \mathbb{E}\sum_S Z_S^2. \tag{A.15}$$

The annular overlap bound [DG21, Lemma 3.16] gives

$$F_S \Longrightarrow C_S \le L, \qquad \sum_S \mathbf{1}_{F_S}C_S \le KL \tag{A.16}$$

for large K. At boundary squares the first implication still holds: the annular outer disk has diameter less than one, so at least one endpoint of a unit-square crossing lies outside it. The second uses overlap multiplicity $O(K^{2\varepsilon_0}) \le K$. Equations (A.12) and (A.16) imply

$$\mathbb{E}\left(\sum_S Z_S\right)^2 \le K^2 \mathbb{E} M_K^2 \le e^{CK^\beta}. \tag{A.17}$$

Split $Z_S = G_S + B_S$ according to $\mathcal{E}$ and its complement. On $F_S \cap \mathcal{E}$,

$$e^{\xi H_S} \le 2^{(q+\zeta)K} e^{CK^\beta}, \qquad A_S/L \le 2^{-(q-\zeta)K} e^{CK^\beta} e^{\xi H_S}.$$

The Gaussian H_S has variance $K \log 2$ and is independent of $\mathcal{G}$. For $X \sim N(0,R)$ and $u \ge 0$,

$$\mathbb{E}[e^{\xi X} \mathbf{1}_{X \le uR}] \le e^{R[\xi \min\{\xi,u\} - \min\{\xi,u\}^2/2]}.$$

The exponent as a function of u is ξ-Lipschitz. Therefore

$$\max_S G_S \le 2^{-(\alpha_* - 2\zeta)K} e^{CK^\beta}, \qquad \alpha_* := \min_{\xi \in I} \begin{cases} q(\xi) - \xi^2/2, & \xi \le Q(\xi), \\ Q(\xi)^2/2, & \xi > Q(\xi) \end{cases} > 0.$$

For the bad part retain the visit indicator: by (A.16), $\mathbf{1}_{F_S} A_S/L \le M_K$. Hence

$$\mathbb{E} \max_S B_S^2 \le \mathbb{E}[\mathbf{1}_{\mathcal{E}^c} M_K^2] \le C e^{-cK^\nu / \log K + CK^\beta}. \tag{A.18}$$

Using $\sum Z_S^2 \le (\max Z_S) \sum Z_S$, Cauchy–Schwarz, and $\zeta = \alpha_*/8$ in (A.15), we obtain

$$\operatorname{Var}(\mathbb{E}[\log L \mid \mathcal{G}]) \le N_{I,A} e^{-c_I K}.$$

Conditional Gaussian concentration for the remaining coarse field gives $\mathbb{E} \operatorname{Var}(\log L \mid \mathcal{G}) \le M_I K$. Thus (A.9) implies

$$\operatorname{Var} \log \widetilde{L}_n \le M_I K + N_{I,A} e^{-c_I K}. \tag{A.19}$$

To close the induction choose R_I with $\mathbb{P}[bZ_U > R_I] < p/4$. Quantile comparison and Chebyshev give

$$\Lambda_n(p) \le \exp\left\{2R_I + C_p \max_{j \le n} \sqrt{\operatorname{Var} \log \widetilde{L}_j}\right\}.$$

Choose A larger than the constant in (A.5) and $C_p\sqrt{M_I + 1} + 2R_I + 1$, then fix a large K for which all preceding estimates hold and $N_{I,A} e^{-c_I K} \le K$. For $n \le K$ the bound $\Lambda_n(p) \le e^{A\sqrt{K}}$ follows from concentration. If it holds below $n > K$, apply (A.19) at each $j \in (K, n]$, using its already proved hypothesis at $j - K$. The last quantile inequality proves the same bound at n. This gives (A.1); substitution in (A.6) proves the annular tails. □

A.2 Passing to the fixed-parameter continuum metrics

The field coupling [DG21, Lemma 4.10] has, on a fixed bounded window, error

$$\mathbb{P}[\|\Phi_{0,n} - h^*_{2^{-n}}\|_U > t] \le C_U e^{-c_U t^2} + o_n(1).$$

Its construction uses only Gaussian fields and geometry, so its error is independent of ξ. Multiplication by $\xi \le b$, followed by the uniform crossing tails, shows that the heat-kernel crossing median $\mathfrak{a}_{2^{-n}}(\xi)$

and $\lambda_n(\xi)$ are comparable with compact-uniform constants. Indeed choose a field-error cutoff and two crossing quantiles whose three exceptional probabilities sum to less than $1/2$. Their product bounds the new median. The same coupling transfers (A.2)–(A.3) to heat-kernel LFPP, with an error tending to zero.

For completeness, here is the deterministic buffered transfer used at the continuum limit. Suppose $d_n \to d$ in the lsc topology, where d is a positive lsc metric, and choose nested annuli $A' = A_{r',R'}(w) \Subset A = A_{r,R}(w)$. Along a subsequence suppose their crossing and loop costs converge to $a, b \in (0,\infty)$. Recovery sequences for points on the two boundary circles of A must cross A', so

$$d(\text{across } A) \geq a, \qquad d(\text{around } A) \leq b. \tag{A.20}$$

For the second assertion parametrize approximate d_n-loops in A' by metric arclength on $[0,1]$. They are Euclidean equicontinuous: on the compact set $\overline{A}'$, lsc convergence and positivity of d give a common eventual positive lower bound for $d_n(x,y)$ when $|x-y| \geq \varepsilon$. Arzelà–Ascoli yields a uniform limit. It still separates A: a compact connecting path disjoint from its trace would have positive Euclidean distance from the trace and hence avoid all late approximating loops. For every finite time partition, the lsc-liminf inequality bounds the limit's length by b.

For each fixed ξ, the established LFPP convergence and uniqueness identify the metric limit as D_h^ξ [DG22, Theorem 1.3][Kav25, Theorem 1.4]. The annular observables can be included in a joint subsequence by their tightness. Apply (A.20) to obtain a uniform lower logarithmic tail for across-costs and upper logarithmic tail for around-costs. The complementary tails follow from finite planar constructions: every separating trace crosses an annulus from a finite small-annulus cover, whereas loops in a fixed chain of overlapping annuli contain a crossing. This proves (3.1), initially without the factor $m(\xi)^{-1}$. Exact scaling gives the normalization $r^{q(\xi)}e^{\xi h_r(w)}$; the field $h(w + r\,\cdot) - h_r(w)$ has the standard pinned whole-plane law. No independence from $h_r(w)$ is needed.

A.3 Moments, access, and confinement

Put $\mu = \min_I q > 0$, $\nu = \max_I q$, and choose $0 < p \leq \min\{1/2, \mu/(4b^2)\}$. Around a deterministic point z, take loops at scales 2^{-k} and paths in wider annuli joining successive loops. The construction in [DG21, Lemma 4.5] uses a bounded number of annular costs at each scale. If $Y_k(z)$ is their sum, including the whole loop, annular moments, scaling, and Cauchy–Schwarz give

$$\mathbb{E}Y_k(z)^p \leq C2^{-pq(\xi)k}(\mathbb{E}e^{2p\xi h_{2^{-k}}(z)})^{1/2} \leq C2^{-3p\mu k/4}. \tag{A.21}$$

Here $\operatorname{Var} h_{2^{-k}}(z) \leq k\log 2 + C_z$. The chain stays in $B_{C2^{-k}}(z)$ and has summable length. Its Euclidean endpoint is z; lower semicontinuity bounds the distance to that endpoint by the remaining tail length. It therefore extends to a metric-continuous path. Connecting two such chains by finitely many fixed-scale annuli inside U proves the positive moment in (3.2). A fixed annulus separating z, w proves all inverse moments. These raw-metric bounds give $\sup_I \mathbb{E}|\log D_h^\xi(0,1)| < \infty$, hence bound both m and m^{-1}. All moment and annular estimates now apply to D^ξ.

To prove Lemma 3.2, let $S_K = \sum_{k \geq K} Y_k(z)$. Since $p \leq 1$, (A.21) gives $\mathbb{E}S_K^p \leq C2^{-3p\mu K/4}$. For $2^{-K_0} \asymp u$, Markov's inequality shows $S_{K_0} \leq v$ except with probability $Cv^{-p}u^{3p\mu/4}$. Every $r \leq u$ uses a loop at a dyadic scale in $[r, 2r)$ and the corresponding chain tail, both of length at most S_{K_0}. The loop lies in $A_{r,4r}(z)$ and the whole construction in $B_{Cr}(z) \subset B_{r^\alpha}(z)$ for small u.

For boundary access, rescale to a disk $B_r(0)$ with $w \in \partial B_r$. Put $\varepsilon_k = 2^{-k}r$ and choose loops in the annuli with radii $(3\varepsilon_k/4, 4\varepsilon_k/5)$ centered at $(1-\varepsilon_k/r)w$ and $(\varepsilon_k/r)w$. Both chains lie inside the disk and share their first loop. Successive centers are $\varepsilon_k/2$ apart. Their inner balls overlap but their filled hulls cannot be nested: containing the next inner ball would require radius at least

$\varepsilon_k/2 + 3\varepsilon_k/8 > 4\varepsilon_k/5$; reverse containment is also impossible. Thus successive traces intersect. For either loop length L_k, scaling and the circle-average increment variance give

$$\mathbb{E}[(S_\xi(0,r)^{-1}L_k)^p] \le C2^{-k(pq(\xi)-p^2\xi^2)} \le C2^{-3p\mu k/4}.$$

The two chains extend to w and 0 as above, proving (3.3). Stop at the half-radius circle, integrate over w, and use Markov's inequality to obtain its bad-arclength estimate.

For the inverse-Hölder estimate choose $B > 2$ and $d > 0$ so that $\chi - q(\xi) - B\xi \ge d$ throughout I. At scale 2^{-k}, choose $O_R(4^k)$ annuli around a fine grid in B_{R+1}, so that some annulus separates any pair at distance comparable to 2^{-k}. Gaussian tails and (3.1) imply, for $t \ge 1$,

$$\begin{aligned}\mathbb{P}[\exists k, w:\ h_{2^{-k}}(w) < -Bk\log 2 - t/(2b)] &\le C_R e^{-ct},\\ \mathbb{P}[\exists k, w:\ X_{k,w} < e^{-t/2}2^{-dk}] &\le C_R e^{-ct},\end{aligned}$$

where $X_{k,w}$ is the normalized crossing cost. In the first sum, $B^2/2 > 2$ beats the grid entropy; in the second, the summands are $C_R 4^k \exp\{-c(t/2 + dk\log 2)^2\}$. Outside these events every selected crossing is at least $ce^{-t}2^{-\chi k}$. A connecting path must cross one of them. This proves (3.4) for $|u - v| \le 1$. For more separated points, stop a finite path on first leaving $B_{1/2}(u)$ and apply the estimate in B_{R+1}, absorbing 2^χ in H.

Finally let L_R be twice a source around-cost in $A_{2R+2,3R+3}$ and C_W its across-cost in $A_{W/2,W}$. Scaling and Cauchy–Schwarz give

$$\mathbb{E}L_R^p \le C(R+1)^{p\nu+p^2b^2}, \qquad \mathbb{E}C_W^{-p} \le CW^{-p\mu+p^2b^2} \le CW^{-3p\mu/4}.$$

If a geodesic with endpoints in B_R leaves B_W, its first and last intersections with a loop of length at most L_R bracket a crossing of $A_{W/2,W}$. Geodesicity gives $C_W \le L_R$. Split this event at $W^{\mu/4}$ and apply the two moment bounds. This proves (3.5), completing the proof of Proposition 3.1.

B The geometry of the annular well

We give the planar constructions used in the proof of Proposition 4.3. All distances in this appendix are Euclidean. The construction is a quantitative version of the annular well of [DG22, Section 5.4].

B.1 A plateau with controlled tubes

Fix $\alpha \in (3/4, 1)$, write $d_0 = 1 - \alpha$, and suppose

$$0 < t \le s \le \min\{\sqrt{t}, d_0/200\}, \qquad 0 < \lambda < c, \qquad 0 < \rho < c\lambda.$$

Let Z consist of $K = \lceil \lambda/(8\rho)\rceil$ equally spaced points of $\partial B_2(0)$, in counterclockwise order. At each $z \in Z$ place a rotated and translated copy, at scale ρ, of a half-annulus with outer radius $R \in [3/4, 1]$ and inner radius αR. Its two gate centers lie on its circular sides, at distance at least $R/3$ from its straight cutline. Each gate has radius $s\rho$. The same construction works if these data vary between chambers. Put $w = \lambda t\rho$.

Lemma B.1 (Plateau construction)**.** *There are smooth simple paths joining the exposed tip of each outer gate to the exposed tip of the next inner gate with the following properties. Their union with the half-annuli and gate disks, thickened by w along the joining paths, is the closure of an annulus $U \Subset A_{3/2,5/2}(0)$ which winds once around 0. Every joining path stays at distance at least $s\rho/2$ from its incident half-annuli. Distinct joining paths, and a joining path and each nonincident chamber or*

gate, have distance at least $d_0\rho/4$. Each path has an embedded tubular neighborhood of radius $ct\rho$, which we may assume is at least $100w$.

The total boundary length, the number of smooth pieces, the curvature bounds, and the reciprocals of the nonzero intersection angles are bounded by a fixed polynomial in $\rho^{-1}, t^{-1}, \lambda^{-1}, d_0^{-1}$. For $0 < a < cw$, there is a simple polygonal cycle $J \subset U$, of winding number one around 0, such that

$$\operatorname{dist}(J, U^c) \geq 10\delta_0, \qquad \sup_{x \in \partial_{\mathrm{out}} U} \operatorname{dist}(x, J) \leq a/4, \qquad \delta_0 \geq caw^C. \tag{B.1}$$

It can be chosen on a square grid of mesh $3\delta_0/2$, with at most $C\delta_0^{-2}$ edges. Here and below the constants may depend on the fixed α.

Proof. First take $R = 1$ and use coordinates in which the half-annulus is $\{\alpha \leq |\zeta| \leq 1,\ \Re\zeta \geq 0\}$. Write its gate centers as $\alpha e^{i\theta_-}, e^{i\theta_+}$, with $\cos\theta_\pm \geq 1/3$. Fix a smooth nondecreasing function χ which is zero on $(-\infty, 0]$, one on $[1, \infty)$, and has derivative at most $3/2$.

From the exposed inner tip $(\alpha - s)e^{i\theta_-}$, go normally inward to radius $\alpha - 2s$, follow this circle counterclockwise to angle π, and then follow the negative real ray to -3. Round the two right-angle joins inside balls of radius $s/100$ using a fixed smooth profile. The circular portion has angular length bounded below by a constant, so the rounding balls are disjoint and leave an initial normal segment of length at least $s/2$. From the exposed outer tip use the polar graph

$$r \exp\{i\theta_+[1 - \chi(r - 3/2)]\}, \qquad 1 + s \leq r \leq 3. \tag{B.2}$$

It initially follows the outward normal and agrees with the positive real ray for $r \geq 5/2$. These two arms have distance at least $s/2$ from the half-annulus. Before the inner arm reaches the negative ray, the radii of the two arms differ by at least d_0; the rounding changes this by at most $s/100$. The negative escape ray and the outer arm have opposite signed real coordinates. Thus the arms are separated by at least $d_0/2$, and have this same clearance from the other gate.

Let n be the positive real direction in the chamber coordinates, let T be the counterclockwise tangent to ∂B_2 at z, and choose $\Delta \in [-\pi, \pi]$ with $e^{i\Delta} n = T$. Continue the two arms through the collar $3 \leq r \leq 6$ by

$$\zeta_\pm(r) = \pm r \exp\{i\Delta\chi((r - 3)/3)\}, \tag{B.3}$$

and then radially to $\pm 8e^{i\Delta}$. The angular derivative is at most $\pi/2 < 2$. If two points on opposite arms have radial coordinates differing by at least $d_0/2$, their distance is at least $d_0/2$. Otherwise their angles differ by at least $\pi - d_0$, so their distance is greater than 5. This argument also handles points on opposite sides of either join at radii 3 and 6: in the only case with small radial difference, the endpoint angle changes by at most twice that difference. The remaining pairs have radial gap at least $d_0/2$. The two entire arms therefore retain the required separation. The interpolation intervals in (B.2)–(B.3) have fixed lengths even when one of their angles is zero.

Scale the construction by ρ. To join the actual circle, write $z = 2e^{i\phi}$, $N = e^{i\phi}$, $T = ie^{i\phi}$, and set

$$\Phi_z(xT + yN) = (2 + y)e^{i(\phi + x/2)} - z, \qquad \Psi_z(v) = v + \beta(|v|/\rho)(\Phi_z(v) - v),$$

where β is a fixed smooth cutoff, zero on $[0, 3]$ and one on $[4, 9]$. On $B_{9\rho}$, $\|D\Psi_z - \mathrm{Id}\| \leq C\rho$ and $\|D^2\Psi_z\| \leq C$. For small fixed ρ, this is a diffeomorphism with distance distortion between $9/10$ and $11/10$. It fixes the chamber and its gates, while mapping the terminal tangent rays to the circle arcs of the same arclength. Join consecutive chambers along the remaining arcs of ∂B_2, from arclength 8ρ after one center to arclength 8ρ before the next. The chamber spacing is greater than 100ρ. The nonterminal arm portions have tangential coordinate at most $6\rho + O(\rho^2)$; the remaining circle

arcs start at 8ρ. This gives a gap of at least ρ from the opposite incident circle arc, and from all nonincident chambers. The two incident arms have gap at least $9d_0\rho/20$.

For $R \in [3/4, 1]$, perform the chamber construction at scale $R\rho$ with normalized gate radius s/R. The physical gate radius remains $s\rho$. The incident-arm gap is at least $9d_0R\rho/20 \geq 27d_0\rho/80 > d_0\rho/4$, and their clearance from the half-annulus is still $s\rho/2$. This is only a Euclidean change of coordinates; no field is repinned.

The curved pieces have curvature at most $C/(s\rho)$. They also have embedded tubular neighborhoods of radius $cs\rho$. For the monotone radial graphs, arc length between radii r, r' is at most $C|r - r'|$, hence at most C times the corresponding chord. Nearby points therefore belong to a short arc, where the curvature bound makes normal coordinates injective; separated arcs have the gaps just proved. The inner circular and radial pieces have the same property, including their fixed-profile roundings. The smooth circle joins preserve it. Decreasing the fixed upper bound for λ gives $ct\rho \geq 100w$. Each arm has boundedly many pieces and length $O(\rho)$; each joining circle arc has length $O(\rho/\lambda)$. Every chamber replaces a circle segment inside a disjoint disk $B_{9\rho}(z)$ avoiding 0, so the resulting cyclic union winds once around 0.

We verify the last assertion by eroding the pieces separately. In cyclic order, write the closed plateau as $P_1 \cup \cdots \cup P_L$, where the pieces are half-annuli, gate disks, and closed tubes. For $0 < \ell < cw$, put

$$K_\ell = \bigcup_i \{x \in P_i : \operatorname{dist}(x, P_i^c) \geq \ell\}.$$

Every piece has an interior cone at scale cw. Consecutive pieces overlap in a disk containing a ball of radius cw; nonconsecutive pieces are separated by at least cw, and there are no triple overlaps. To check the overlaps, a gate centered on a circular side cuts off a cap of depth comparable to its radius, away from the straight cutline. A tube ends normally at the exposed gate tip, and its terminal radius-w disk cuts off a cap of depth w. After erosion these depths decrease by at most 2ℓ. The overlap boundaries remain transverse. The first type has an angle bounded below by a constant; for the second, the angle is bounded below by a fixed power of $w/(s\rho)$. Thus all these statements persist for $\ell < cw$ with a smaller fixed c. Attaching these eroded disks in cyclic order leaves precisely one hole. Consequently K_ℓ is the closure of an annulus and

$$\operatorname{dist}(K_\ell, U^c) \geq \ell, \qquad \sup_{x \in \partial_{\mathrm{out}} U} \operatorname{dist}(x, \partial_{\mathrm{out}} K_\ell) \leq C\ell.$$

For the second inequality, move x into the eroded incident piece along its interior cone, a distance at most $C\ell$. The segment first enters K_ℓ through its outer boundary because x is accessible from the unbounded complementary component.

The eroded boundaries consist of the same finite collection of circular, straight, and offset smooth pieces. Their lengths, curvatures, separation margins, and intersection angles have polynomial bounds. Choose ℓ a small fixed multiple of a, and choose $\delta_0 \geq caw^C$ small enough, by increasing C, that a grid of mesh $3\delta_0/2$ resolves these margins and angles. The outer boundary of the cells meeting the filled hull of K_ℓ lies within $C\delta_0$ of $\partial_{\mathrm{out}} K_\ell$. Removing bounded grid indentations gives a simple grid cycle in the same homotopy class; the separation and angle bounds ensure that this operation changes the boundary only within $C\delta_0$ of the original one. Take $C\delta_0 < \ell/10$ and $C\ell < a/8$. This proves (B.1). A bounded site contains at most $C\delta_0^{-2}$ grid edges. □

B.2 Finding the central gate segments

Write $V = B_\omega(U)$. Suppose $0 < \omega \ll d \ll s\rho$ and $d \ll d_0\rho$. A gate disk of radius $s\rho$ is then enlarged to radius $s\rho + \omega$ in V.

Lemma B.2 (Gate routing)**.** *Let $P : [\tau, \sigma] \to \mathbb{C}$ be a continuous path with endpoints on ∂V, separated by at least a_4, where $\rho \le ca_4$. Suppose each component of its portion outside V has diameter at most d. Mark a chamber in every circle arc of length $a_4/2$. There is a marked chamber, at distance at least 100ρ from both endpoints, such that for each of its two enlarged gates $B = B_{R+\omega}(z_0)$, $R = s\rho$, there are $\tau < a < b < \sigma$ with*

$$\begin{gathered} P(a), P(b) \in \partial B, \qquad |P(a) - P(b)| \ge R/8, \\ P([a,b]) \subset \overline{B}_{R+\omega+d}(z_0) \setminus (V \setminus \overline{B}). \end{gathered} \tag{B.4}$$

The last hit before a and the first hit after b of $\partial B_{R+\omega+100d}(z_0)$ both occur in (τ, σ). No restriction on visits to B is imposed on these larger side portions.

Proof. The plateau lies within 10ρ of ∂B_2. Remove the 100ρ-neighborhoods of the two endpoint projections from that circle. For small enough ρ/a_4, each of the two remaining arcs has length at least $a_4/2$. Select a marked chamber on each, and call their four enlarged gates $B(u), B(v), B(u'), B(v')$. Removing these gates from V leaves four components in cyclic order: an endpoint component V^τ, a chamber component O, the other endpoint component V^σ, and the other chamber component O'. The closures of distinct components have distance at least $R/4$. Indeed, at a gate the chamber exits near the two lateral circle points whereas the tube exits through the exposed normal tip; the tube width is at most λR and $\omega \ll R$. All other gaps follow from the clearances in Lemma B.1. Each component is incident to exactly its two neighboring gate disks.

Let d' be the first arrival in $\overline{V}^\sigma$. Since an excursion outside V cannot span any of these gaps, this arrival is within d of $B(v)$ or $B(u')$. Relabel the two chambers and reverse their cyclic order if needed so that it is near $B(v)$. Before time d' the path must visit O: the components of $V \setminus (V^\sigma \cup O)$ containing V^τ and $B(v)$ are separated by at least $d_0\rho/2 > d$. Let b' be its first arrival in $\overline{O}$. It is within d of $B(u)$, since every other component it could previously visit is farther than d from O.

Let a' be the last departure from $\overline{V}^\tau$ before b'. The endpoints $P(a'), P(b')$ have distance at least $R/4$. Between them the path avoids $V^\tau \cup O$. The d-neighborhood of the remaining components and gates has one component equal to $B_d(B(u))$ and all its other components are separated from it. Continuity, and the diameter bound on excursions outside V, give

$$P([a', b']) \subset \overline{B}_d(\overline{B}(u)) \setminus (V \setminus \overline{B}(u)).$$

Let a and b be its first and last visits to $B(u)$ on this interval. They exist, and each is within d of the corresponding endpoint of the interval; otherwise an excursion outside V would have diameter greater than d. Hence $|P(a) - P(b)| \ge R/4 - 2d \ge R/8$. This proves (B.4) for $B(u)$. Applying the same argument between the last departure from O and the first arrival in V^σ proves it for $B(v)$. Both global endpoints are farther than 100ρ from the selected chamber, while $R + \omega + 100d \ll \rho$. Continuity gives the stated outer bracketing times, strictly inside the original interval. □

B.3 Replacing repeated visits by circle arcs

Lemma B.3 (Circle replacement)**.** *Let $B = B_r(z_0)$, let $u \in B_{r/2}(z_0)$, and let $P : [a, b] \to \mathbb{C}$ be injective and continuous, with distinct endpoints on ∂B, $u \notin P$, and $|P \cap \partial B| = 0$, where $|\cdot|$ denotes circle arclength. Suppose every component of P outside $\overline{B}$ has diameter at most $d < r/100$.*

For each component of $P^{-1}(B)$, let its selected boundary arc be the arc cut off from u by that interior crosscut. Denote their union by $\mathcal{B}$. For each exterior component, select the minor boundary arc between its endpoints, and denote their union by $\mathcal{O}$. Then

$$|\mathcal{B} \cup \mathcal{O}| \ge |P(a) - P(b)|. \tag{B.5}$$

The exterior arcs have length at most $2d$ and form a laminar family. Its maximal members exist, have pairwise disjoint interiors, and have the same union $\mathcal{O}$. Each maximal arc retains its original exterior path segment as a bracket.

Proof. Put $F = P^{-1}(\partial B)$. Replace each component of $[a, b] \setminus F$ by a traversal of its selected circle arc, retaining P on F. The resulting circle-valued map Q is continuous. To check this at an accumulation time $t_0 \in F$, consider distinct component intervals approaching t_0. Their lengths tend to zero, and uniform continuity makes their path images shrink to $P(t_0)$. Exterior minor arcs shrink as well. An interior crosscut sufficiently near $P(t_0)$, together with the short boundary arc between its endpoints, bounds a small Jordan domain disjoint from u. Its selected arc is therefore this short arc and also shrinks to $P(t_0)$. The connected set $Q([a, b])$ contains $P(a), P(b)$, so has arclength at least their Euclidean distance. It is contained in $\mathcal{B} \cup \mathcal{O} \cup (P \cap \partial B)$, which proves (B.5).

An exterior path segment and its minor arc form a Jordan curve in a ball of radius $2d$ about an endpoint. Its bounded complementary component is exterior to B, since this ball does not contain the center of B. If two selected arcs had interlacing endpoints, the exterior path for one would enter and leave the exterior cap of the other, forcing the two path segments to intersect. Injectivity excludes this. Since every selected arc has length less than πr, noninterlacing means that two arcs have disjoint interiors or are nested.

For each $\varepsilon > 0$, only finitely many component intervals have path images of diameter at least ε: uniform continuity gives a positive lower bound for their time lengths, and their interiors are disjoint. Every arc containing a given positive-length arc has an endpoint chord bounded below by a fixed multiple of that length. It therefore has only finitely many strict super-arcs. Every arc belongs to a maximal one, and laminarity proves the remaining assertions. □

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